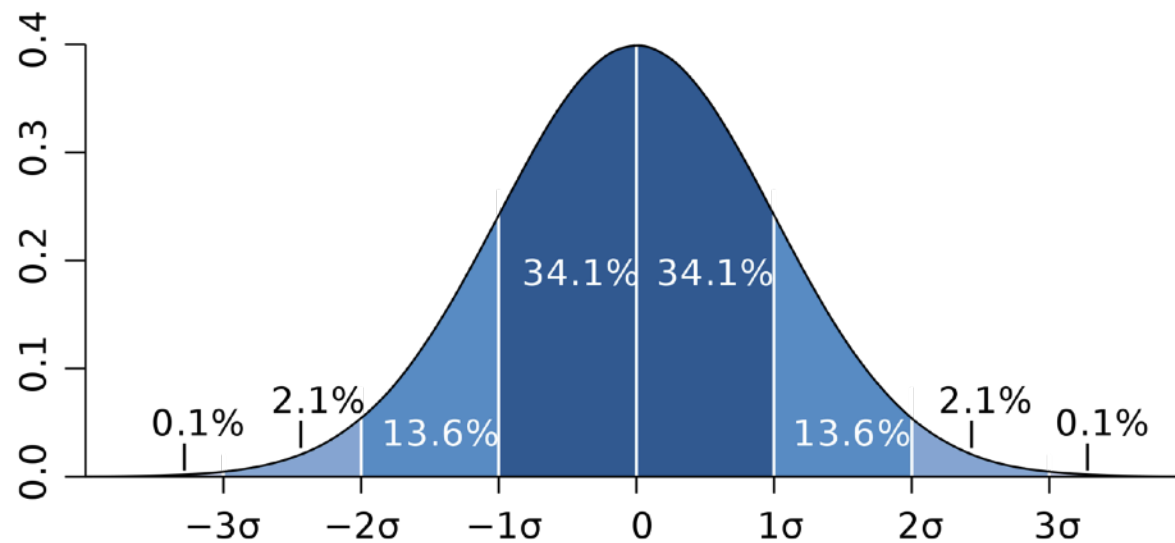




FELIPE ORTEGA

IVÁN MAURICIO BURBANO ALDANA

RATE OF CHANGE, CALCULUS, AND DIFFERENTIAL EQUATIONS

BASED ON SLIDES DESIGNED BY R. BRICEÑO AND T. ROGERS

REVIEW

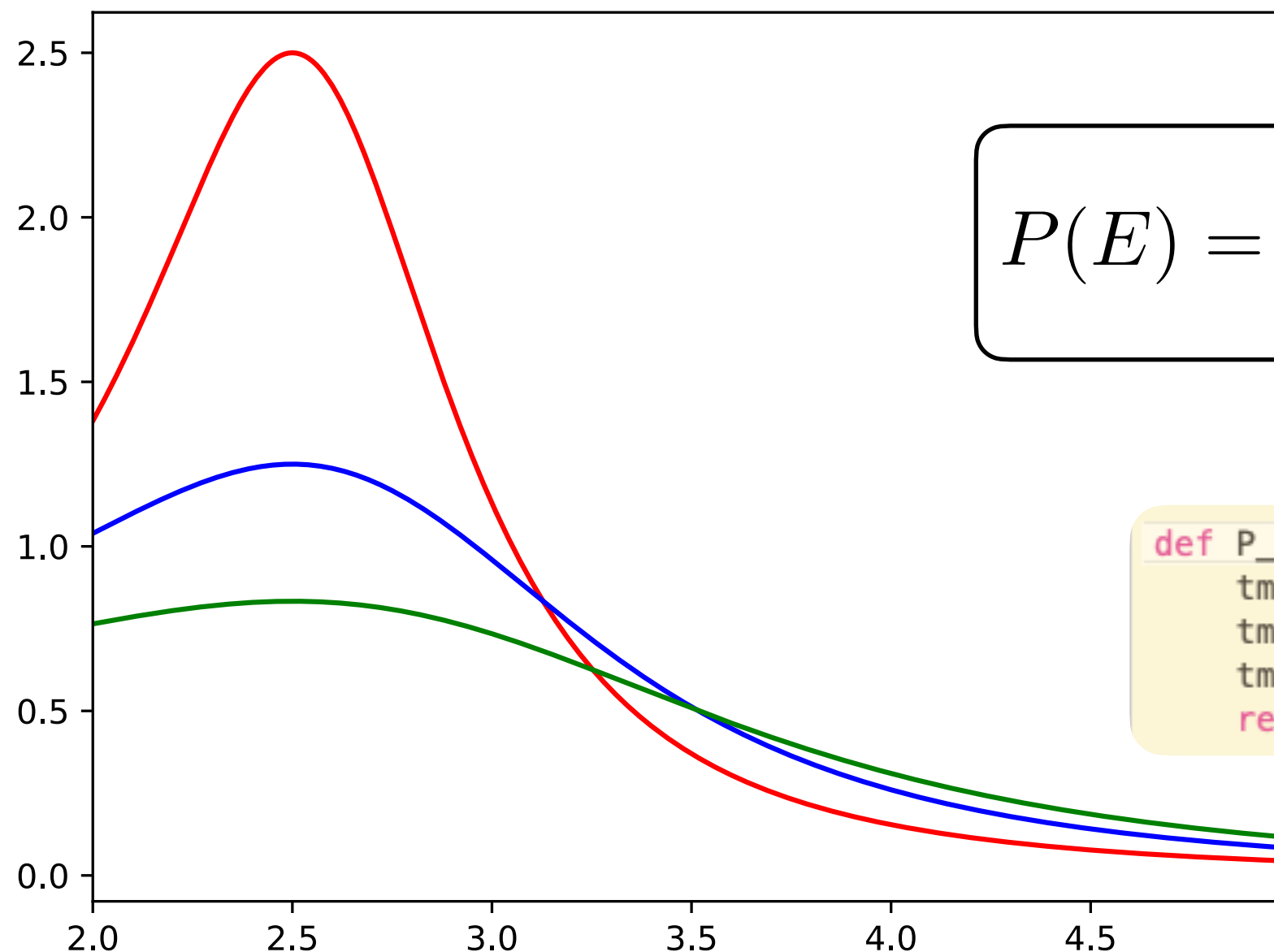


EXERCISE #1 - PLOTTING A BREIT-WIGNER

The three values of the decay width are $\Gamma = 1, 2, 3$.

Reproduce this plot, and figure out which color correspond to which value.

Hint: Develop some intuition. How does the distribution behave as a function of the width at its peak. Does it grow or decrease with the Γ ?



$$P(E) = \frac{M^3 \Gamma}{(E^2 - M^2)^2 + M^2 \Gamma^2}$$

```
def P_BW(E):  
    tmp1 = pow(pow(E,2)- pow(M,2) , 2)  
    tmp2 = pow(M*Gamma,2)  
    tmp3 = pow(M,3)*Gamma  
    return tmp3 / ( tmp1 + tmp2 )
```

look inside my code 😎

PYTHON ROUTINES FOR MINIMIZATION

scipy has many different routines for optimization and root finding: <https://docs.scipy.org/doc/scipy/reference/optimize.html>

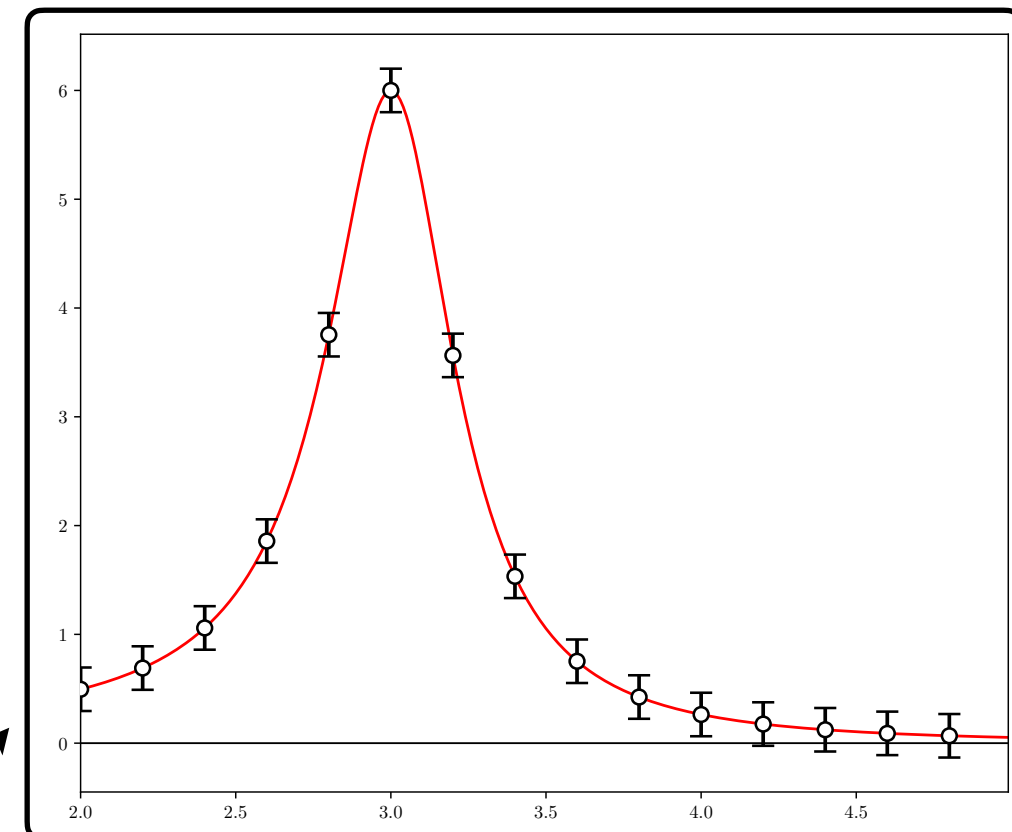
Here is a simple one I use

```
def chi2_BW(M_Gamma):  
    M, Gamma = M_Gamma  
    Pis = P_BW(M, Gamma, Xs)  
  
    chi20 = np.sum(pow((Pis-Ys)/sigs,2))  
    dof = len(sigs)-2.0  
    return chi20  
  
def find_min_chi2_BW():  
    M_Gamma_guess = [2.,1.0]  
    """  
    https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html  
    """  
    from scipy import optimize  
    M_Gamma = optimize.minimize(chi2_BW, M_Gamma_guess, method='nelder-mead')  
    print("M_Gamma", M_Gamma)  
    return M_Gamma.x
```

minimum coincides
with $(M, \Gamma) = (3, 0.5)$

```
In [80]: import BW_plots  
M_Gamma final_simplex: (array([[2.99995863, 0.49998755],  
[2.9999788 , 0.50007577],  
[3.00004901, 0.49990999]]), array([2.17813848e-05, 3.02255241e-05, 6.72629601e-05]))  
      fun: 2.1781384810732297e-05  
      message: 'Optimization terminated successfully.'  
      nfev: 103  
      nit: 55  
      status: 0  
      success: True  
      x: array([2.99995863, 0.49998755])
```

plotting the distribution
using the fitted values...



PYTHON ROUTINES FOR MINIMIZATION

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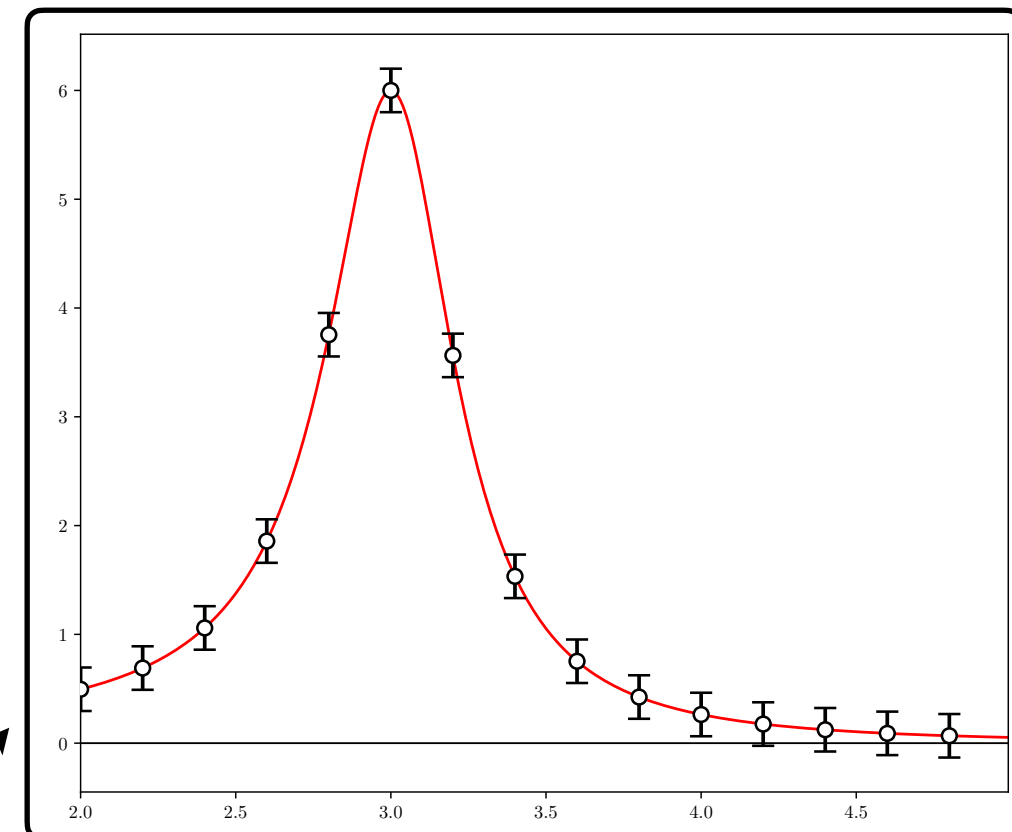
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    return M_Gamma.x
```

minimum coincides
with $(M, \Gamma) = (3, 0.5)$

the resulting values of (M, Γ)

```
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```

plotting the distribution
using the fitted values...



EXERCISE #2 - WARM UP...FIT A CONSTANT

Load the data in [fitting_data/const_data.txt](#)

independent
variable

average measurement

standard deviation

to load, you can see
the following code

```
filename = 'const_data.txt'
const_data = np.loadtxt(filename)
print("shape(const_data) = ", np.shape(const_data))
Xs, Ys, sigs = const_data.T
print("shape(Xs) = ", np.shape(Xs))
print("shape(Ys) = ", np.shape(Ys))
print("shape(sigs) = ", np.shape(sigs))
```

output...

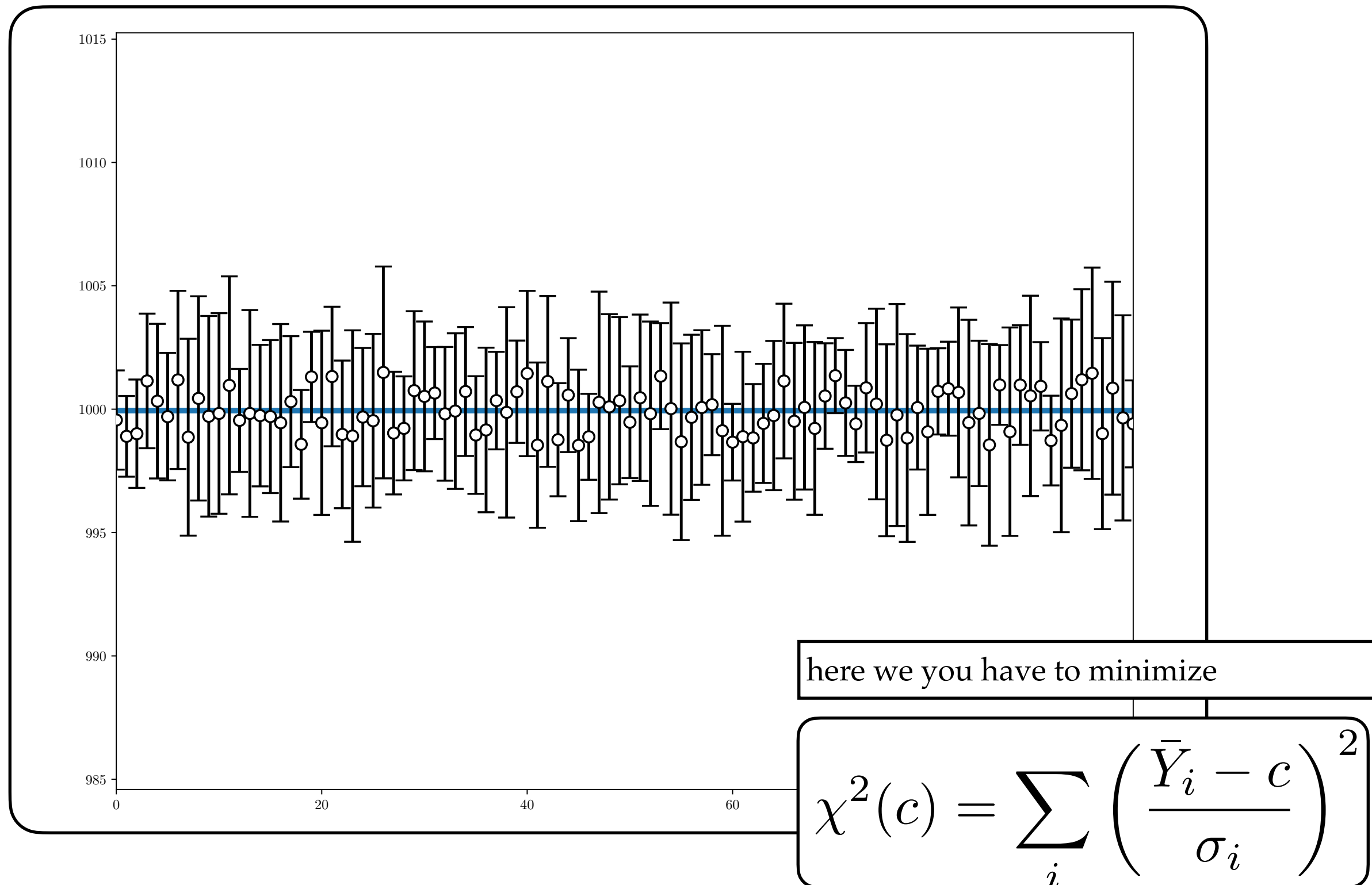
```
shape(const_data) = (100, 3)
shape(Xs) = (100,)
shape(Ys) = (100,)
shape(sigs) = (100,)
```

	independent variable	average measurement	standard deviation
1	0.0000000000000000e+00	9.995608073198061447e+02	2.010880856634203795e+00
2	1.0000000000000000e+00	9.989000304105954910e+02	1.635795591692006834e+00
3	2.0000000000000000e+00	9.990060216805485425e+02	2.196269491332777601e+00
4	3.0000000000000000e+00	1.001145019684710974e+03	2.727748014799645127e+00
5	4.0000000000000000e+00	1.000325729760938771e+03	3.133910015053178011e+00
6	5.0000000000000000e+00	9.996966972703230567e+02	2.582352105425600453e+00
7	6.0000000000000000e+00	1.001185862556614097e+03	3.610132155324399239e+00
8	7.0000000000000000e+00	9.988642623323200951e+02	3.990053363012680876e+00
9	8.0000000000000000e+00	1.000437143482329930e+03	4.138951037265467647e+00
10	9.0000000000000000e+00	9.997121768891632883e+02	4.065402221831392993e+00
11	1.0000000000000000e+01	9.998270261017927396e+02	4.065727850605284743e+00
12	1.1000000000000000e+01	1.000965868351814720e+03	4.418148420465375814e+00
13	1.2000000000000000e+01	9.995462363367937542e+02	2.086260784909980348e+00
14	1.3000000000000000e+01	9.998277159350965348e+02	4.191409335069876008e+00
15	1.4000000000000000e+01	9.997391408396299539e+02	2.868036357268816339e+00
16	1.5000000000000000e+01	9.996997858726948607e+02	3.100581750743879539e+00
17	1.6000000000000000e+01	9.994488473562142872e+02	4.001916818957766075e+00
18	1.7000000000000000e+01	1.000306724780818513e+03	2.653736388738097141e+00
19	1.8000000000000000e+01	9.985763846692332208e+02	2.203381851084773579e+00
20	1.9000000000000000e+01	1.001307855283192339e+03	1.830367272051301120e+00
21	2.0000000000000000e+01	9.994477788298243013e+02	3.733641562551535564e+00
22	2.1000000000000000e+01	1.001323711961507342e+03	2.828454877578798765e+00
23	2.2000000000000000e+01	9.989779317670313503e+02	2.992168487437034230e+00
24	2.3000000000000000e+01	9.989108166430121400e+02	4.285277418619901191e+00
25	2.4000000000000000e+01	9.996794307426688420e+02	2.803714930261759442e+00
26	2.5000000000000000e+01	9.995316088565639348e+02	3.519255602155442642e+00
27	2.6000000000000000e+01	1.001489465965241038e+03	4.293321083141394823e+00
28	2.7000000000000000e+01	9.990306379092597808e+02	2.485797284494362103e+00
29	2.8000000000000000e+01	9.992218459646275051e+02	2.108402591512725799e+00
30	2.9000000000000000e+01	1.000751406739915751e+03	3.220350739074319790e+00
31	3.0000000000000000e+01	1.000517685400707251e+03	3.036311141027919547e+00
32	3.1000000000000000e+01	1.000653489026880834e+03	1.864845433477076986e+00
33	3.2000000000000000e+01	0.008120851437173542e+03	2.700866504021734354e+00

EXERCISE #2 - WARM UP...FIT A CONSTANT - WHAT TO EXPECT

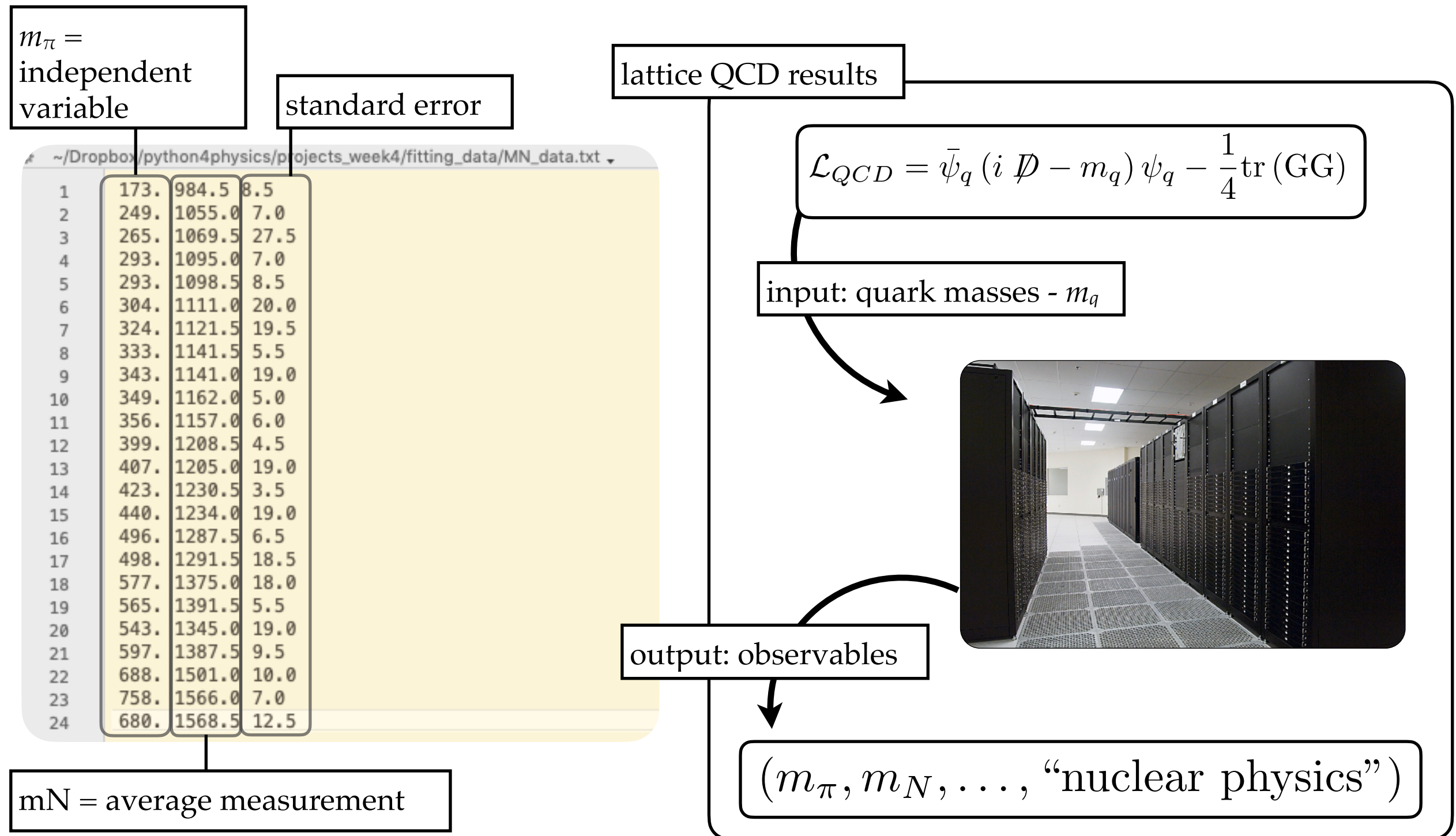
Load the data in [fitting_data/const_data.txt](#)

Fit the data to a constant. Follow up, how does this constant compare to the average?



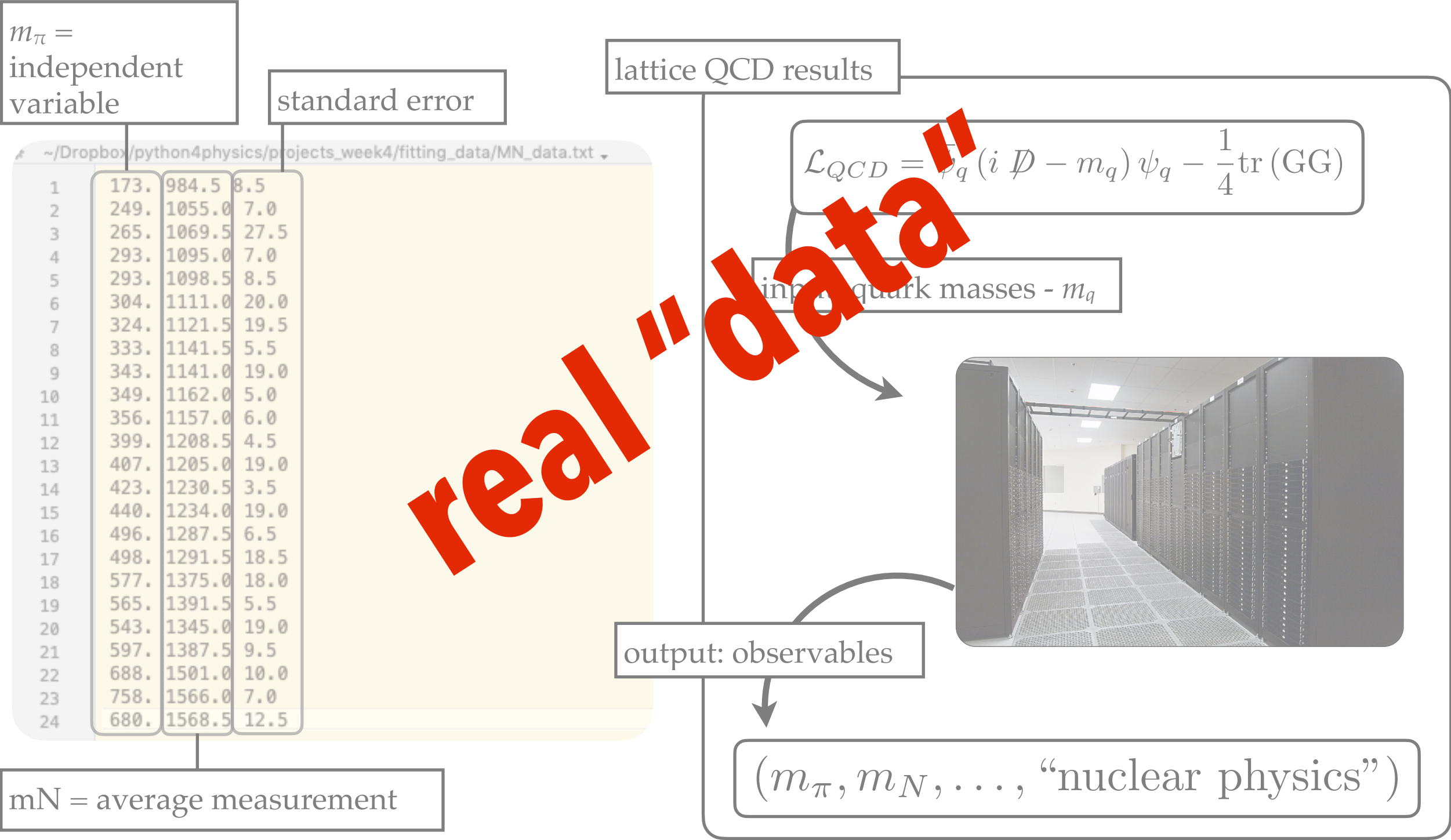
EXERCISE #3 - FIT A STRAIGHT LINE

Load the data in [fitting_data/MN_data.txt](#)



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Load the data in [fitting_data/MN_data.txt](#)

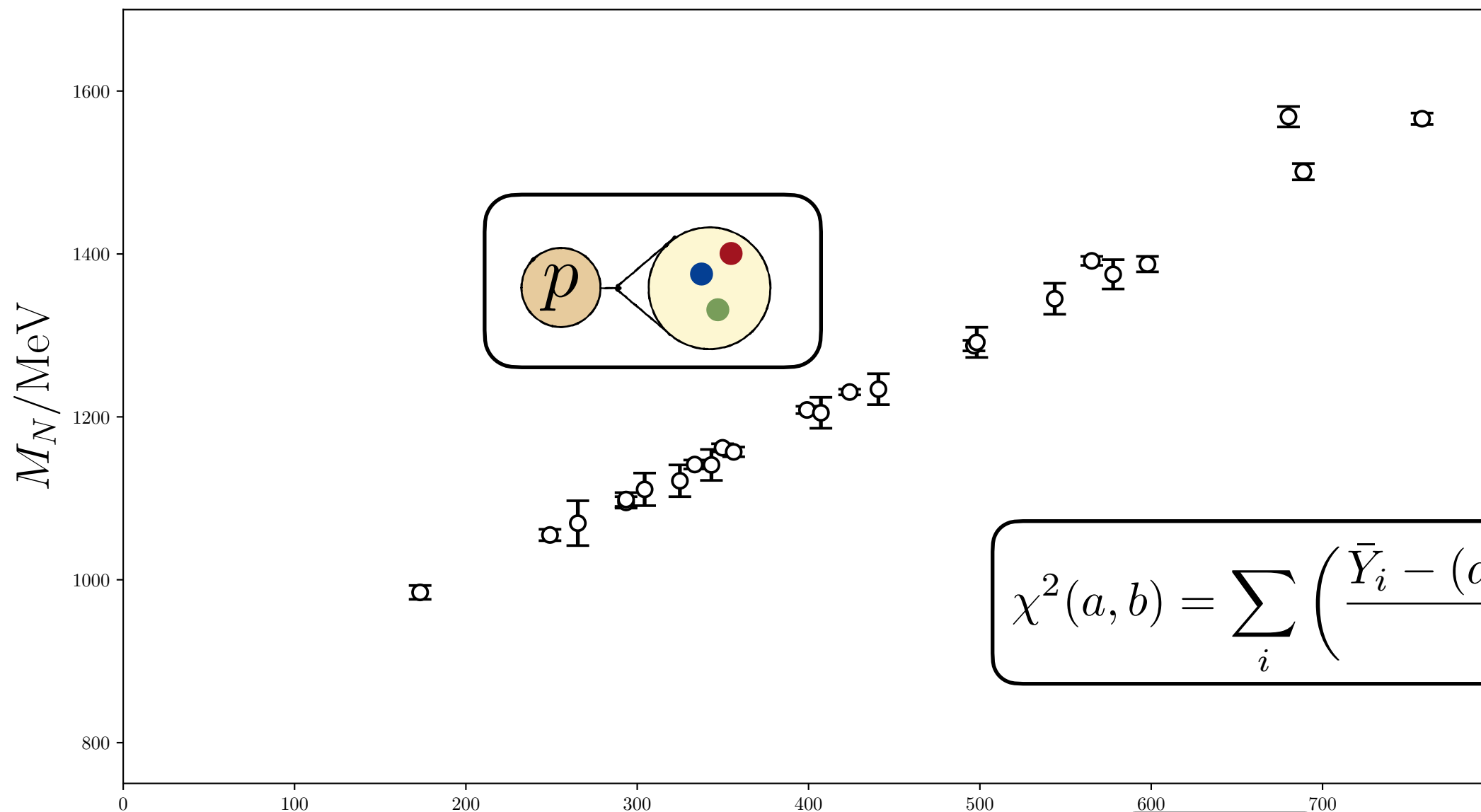


EXERCISE #3 - FIT A STRAIGHT LINE

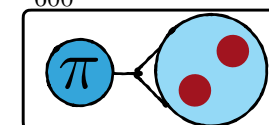
Load the data in [fitting_data/MN_data.txt](#)

Fit to a straight line assuming $M_N = M_0 + m_\pi b$

M_0 is the value of the mass when the Higgs does not matter! Compare this to the physical nucleon mass. How much does the Higgs contribute to our mass?



$$\chi^2(a, b) = \sum_i \left(\frac{\bar{Y}_i - (a + b m_{\pi,i})}{\sigma_i} \right)^2$$



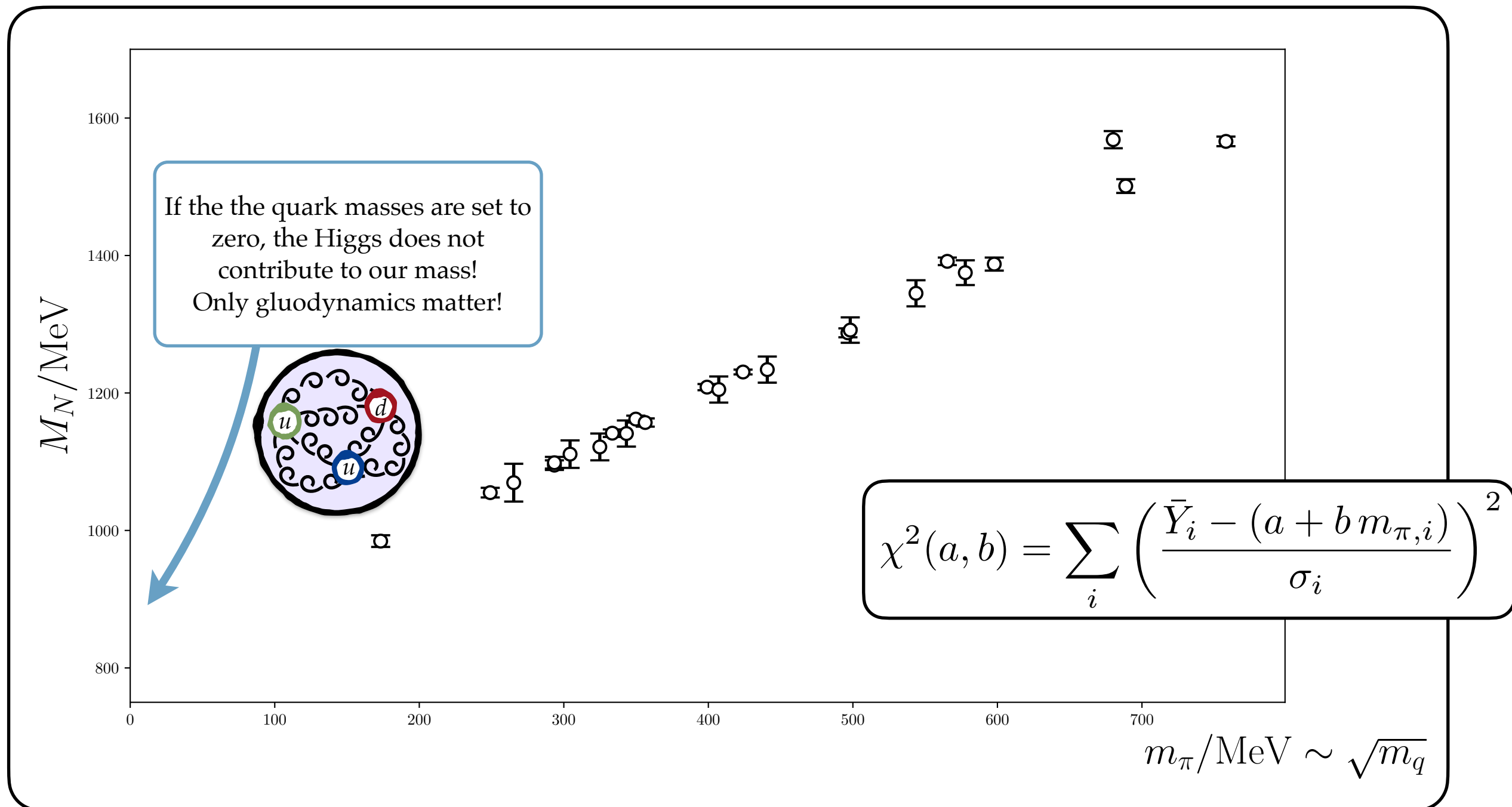
$$m_\pi/\text{MeV} \sim \sqrt{m_q}$$

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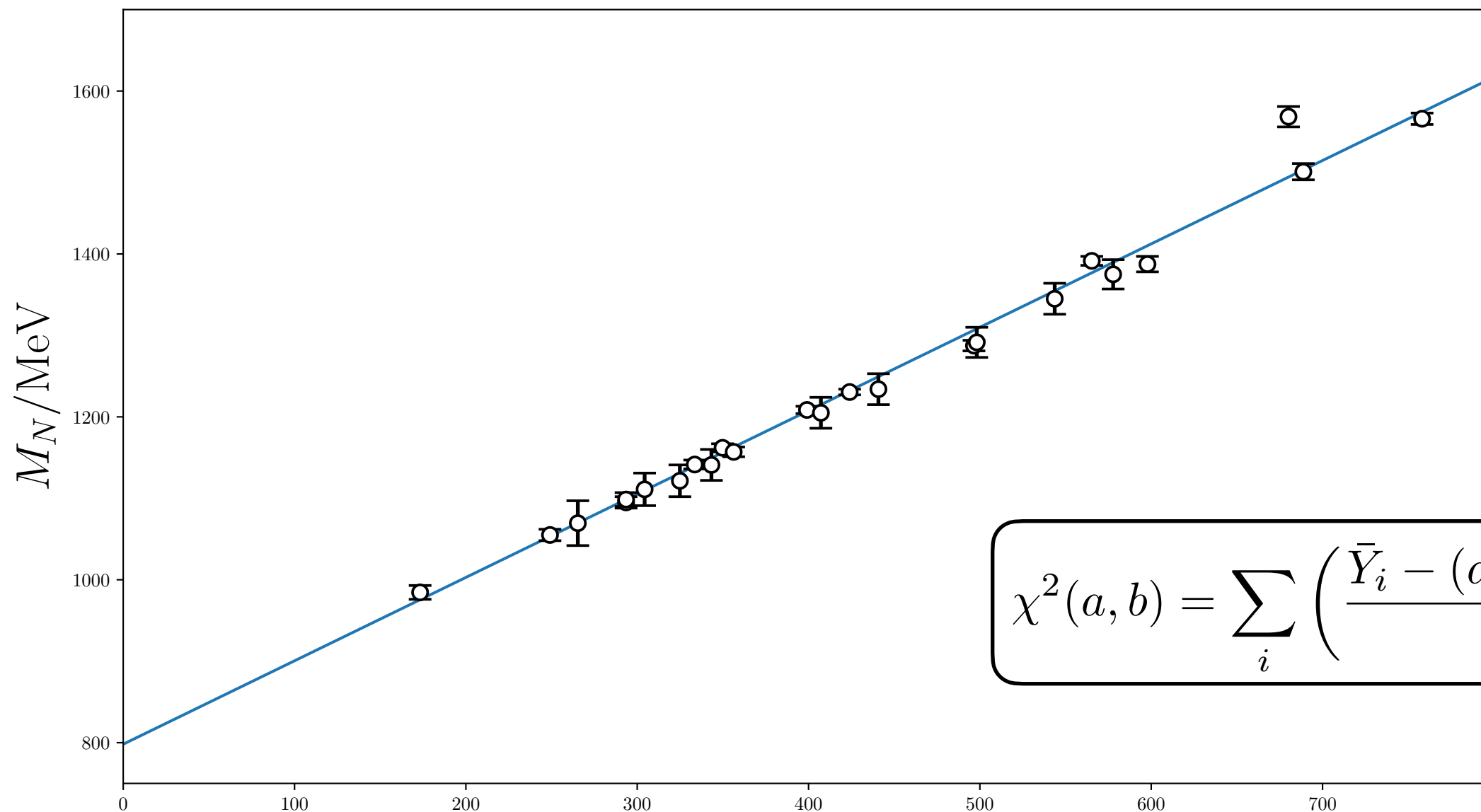


EXERCISE #3 - FIT A STRAIGHT LINE - WHAT TO EXPECT

Load the data in [fitting_data/MN_data.txt](#)

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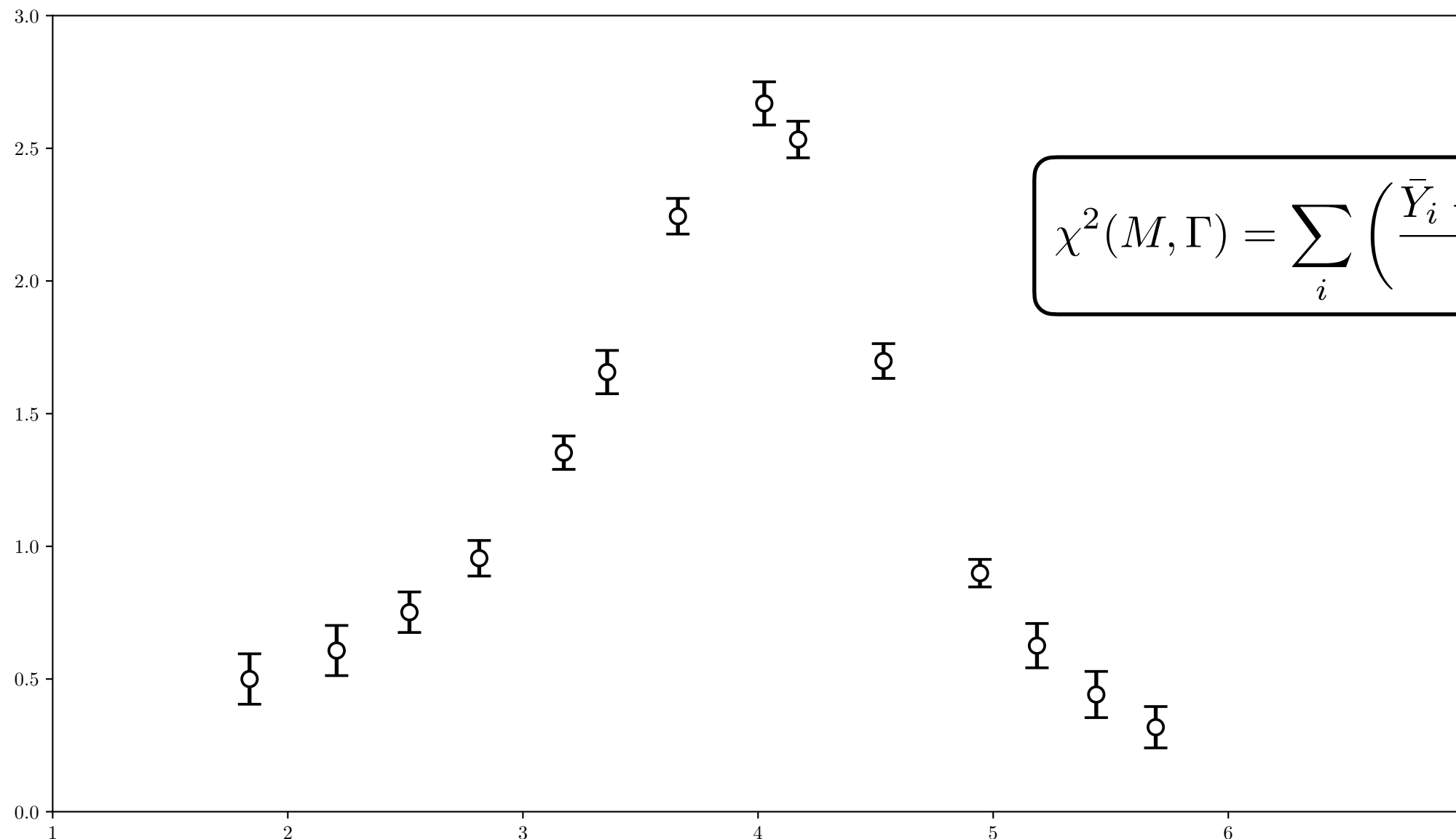
$$\chi^2(a, b) = \sum_i \left(\frac{\bar{Y}_i - (a + b m_{\pi,i})}{\sigma_i} \right)^2$$

$$m_\pi / \text{MeV} \sim \sqrt{m_q}$$

EXERCISE #4 - FIT A BREIT-WIGNER

Load the data in [fitting_data/BW_data.txt](#)

Fit to a Breit-Wigner. Determine the mass and width of the state

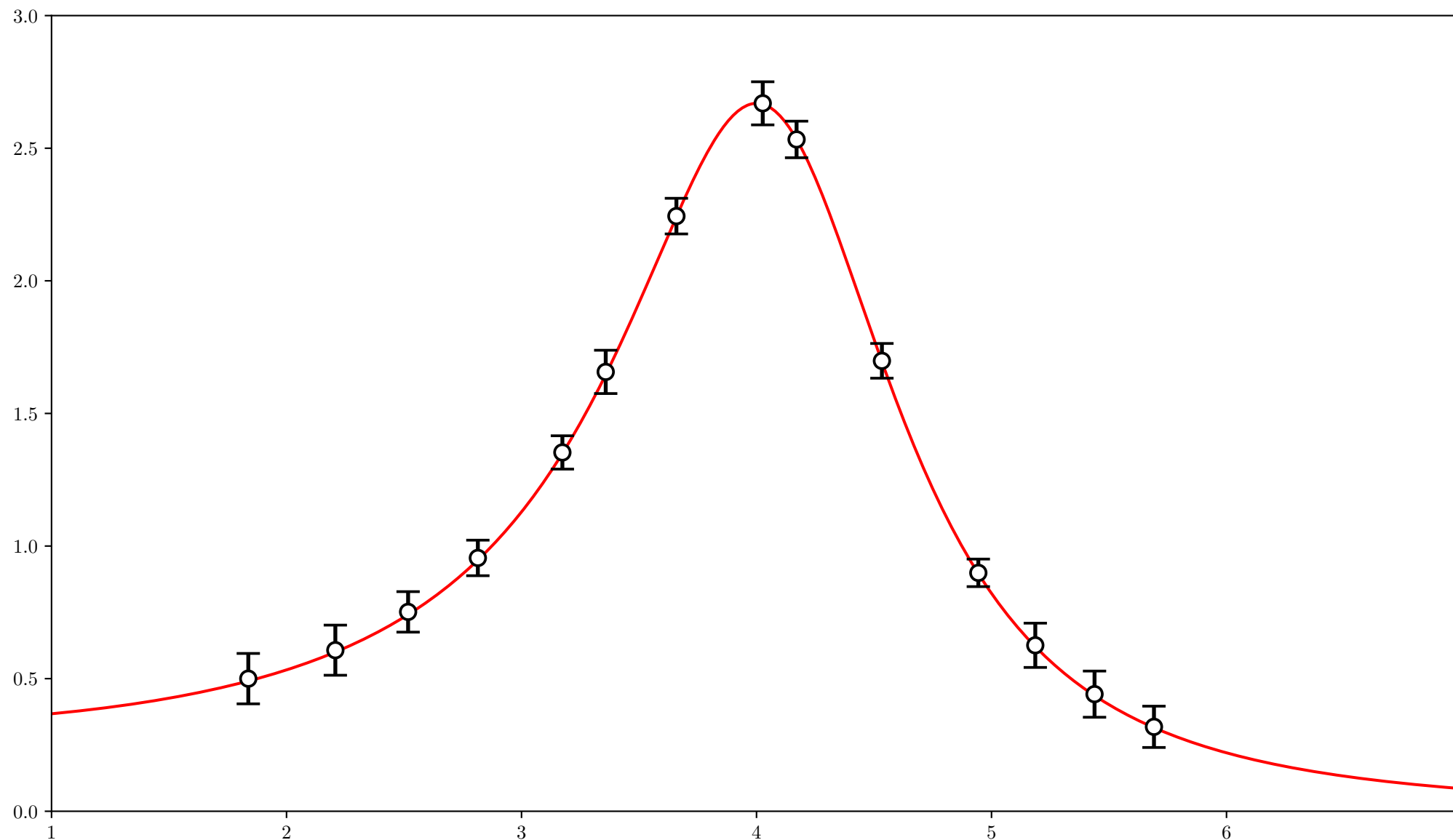


$$\chi^2(M, \Gamma) = \sum_i \left(\frac{\bar{Y}_i - P_i(M, \Gamma)}{\sigma_i} \right)^2$$

EXERCISE #4 - FIT A BREIT-WIGNER - WHAT TO EXPECT

Load the data in [fitting_data/BW_data.txt](#)

Fit to a Breit-Wigner. Determine the mass and width of the state



SUMMARY - EQUATIONS USED

Statistics:

Goodness of fit:
$$\chi^2(\{c\}) = \sum_i \left(\frac{\bar{Y}_i - f(X_i, \{c\})}{\sigma_i} \right)^2$$

X_i = independent variables, Y_i = dependent variables, σ_i = errors in Y_i ,
 f = function of variables $\{c\}$, which represent the expected value.

Physics:

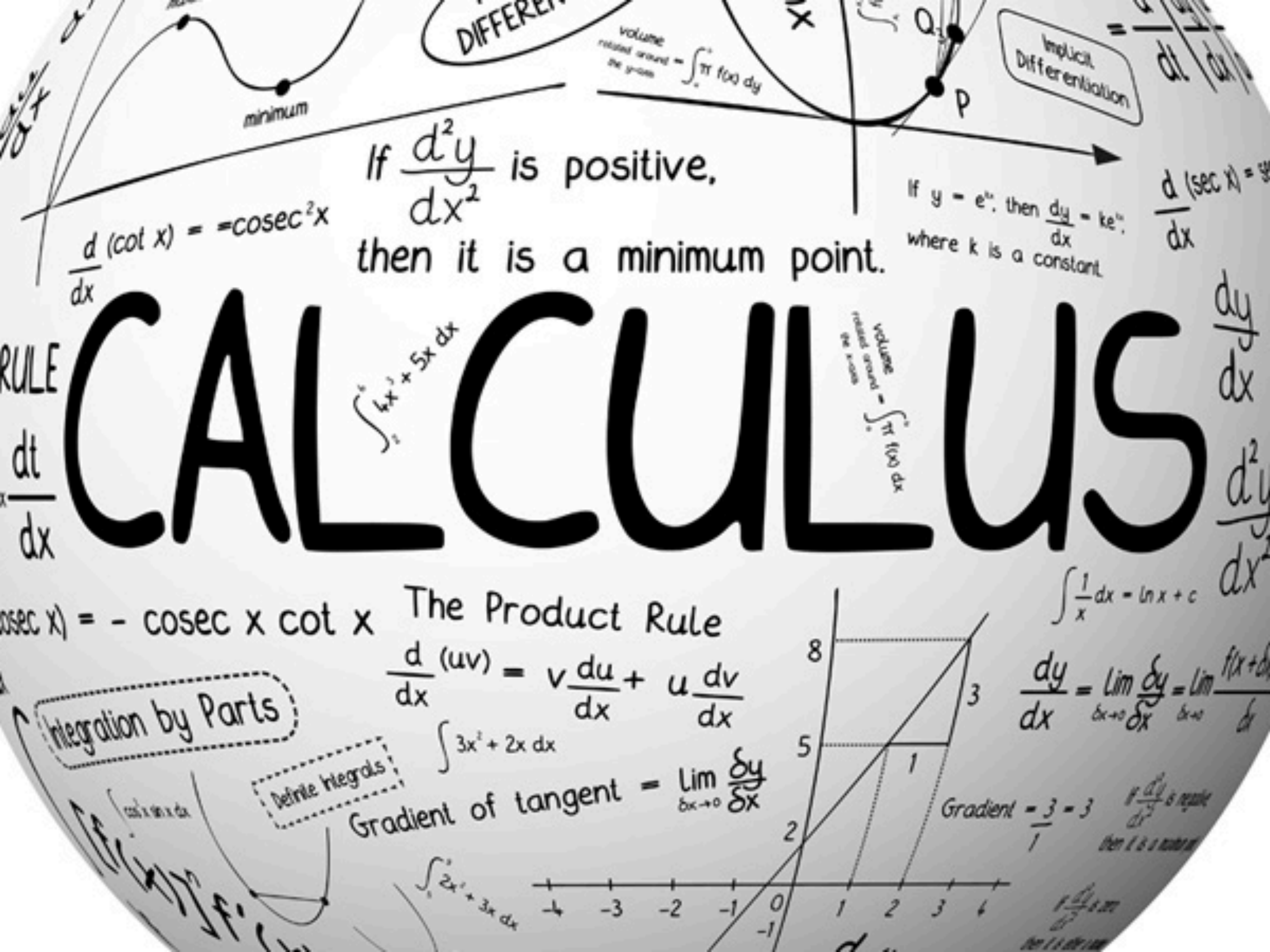
Breit-Wigner distribution as a function of energy (E):

$$P(E) = \frac{M^3 \Gamma}{(E^2 - M^2)^2 + M^2 \Gamma^2}$$

M, Γ are properties of a short-lived resonating particle.

M = is the mass of the resonance

Γ = decay width of the resonance = $1 / (\text{lifetime of the resonance})$



CALCULUS AND INFINITESIMAL DIFFERENCES

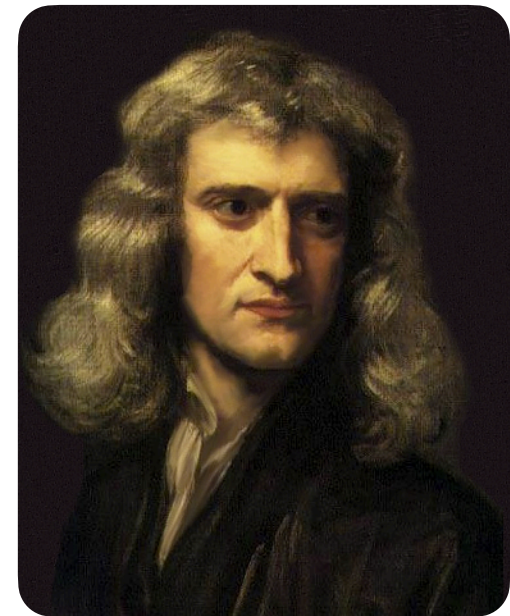
Calculus is a remarkably powerful field of mathematics. Although its history dates back to ancient Greece, the formalization of the field is largely attributed to **Newton** and **Leibniz**, who made important independent contributions.

Leibniz's interest lied primarily in setting foundational notation and mathematical rules. While **Newton's** interest was always describing the motion of physical objects, which is our interest as well.

Calculus can be divided into two fields:

differential calculus: interested in the rate of change of quantities, or equivalently the slopes of the corresponding curves

integral calculus: interested in the continuous sum of function, or equivalent the area under the curve of a function or between different functions.



Isaac Newton

England, 1643 - 1727



Gottfried Wilhelm Leibniz

France, 1646 - 1716

CHANGE AND THE SYMBOL FOR IT

In what follows we are interested in evaluating the change of different quantities.

Consider a function of some independent variable x : $f(x)$

At some point x has a value x_1 . At some later time it has the value x_2 .

We can define the change in x using the Greek letter Δ : $\Delta x \equiv x_2 - x_1$

Similarly, we can define the change in $f(x)$: $\Delta f \equiv f(x_2) - f(x_1)$.

At times we will want to specify the shift in $f(x)$ in a particular way. We will define the initial and final point as $x_1 = x - a$ and $x_2 = x + a$ respectively where the distance from final and initial values of x is $x_2 - x_1 = 2a$. The the difference labeled by a is defined as $\Delta_a f(x) \equiv f(x + a) - f(x - a)$

DESCRIBING MOTION



descriptive question

[*Where is the object?*]

descriptive quantities

[*position*]

DESCRIBING MOTION



descriptive question

Where is the object?

[How far has it travelled?]

descriptive quantities

position

[displacement, distance]

DESCRIBING MOTION



descriptive question

Where is the object?

How far has it travelled?

*[Where is it going?
How fast is it going?]*

descriptive quantities

position

displacement, distance

[speed, velocity]

DESCRIBING MOTION



descriptive question

Where is the object?

How far has it travelled?

Where is it going?

How fast is it going?

[Is it speeding up, slowing down, or neither?]

descriptive quantities

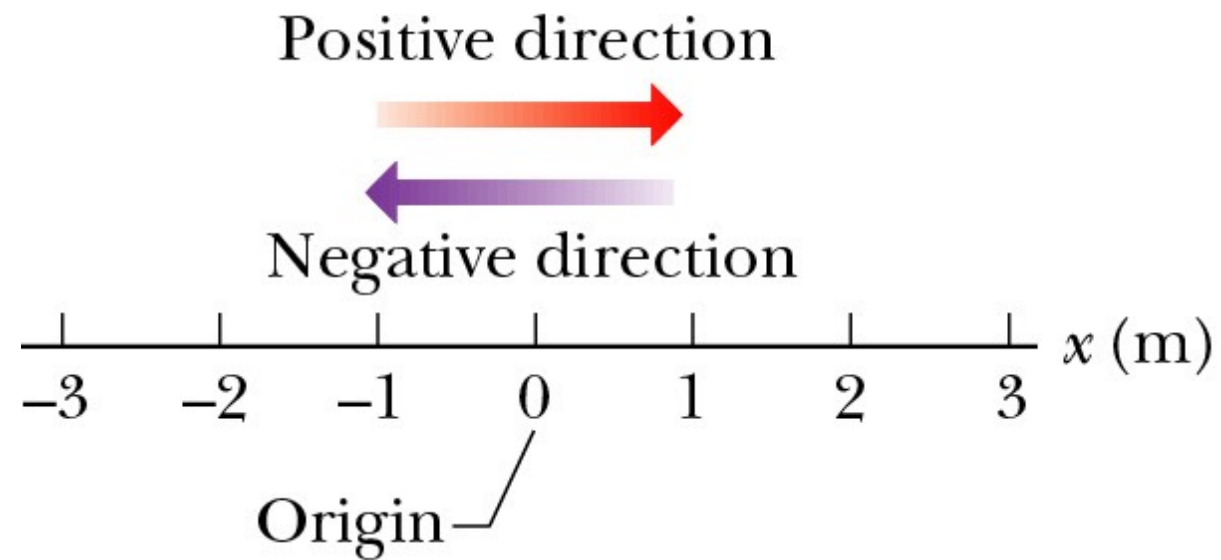
position

displacement, distance

speed, velocity

[acceleration]

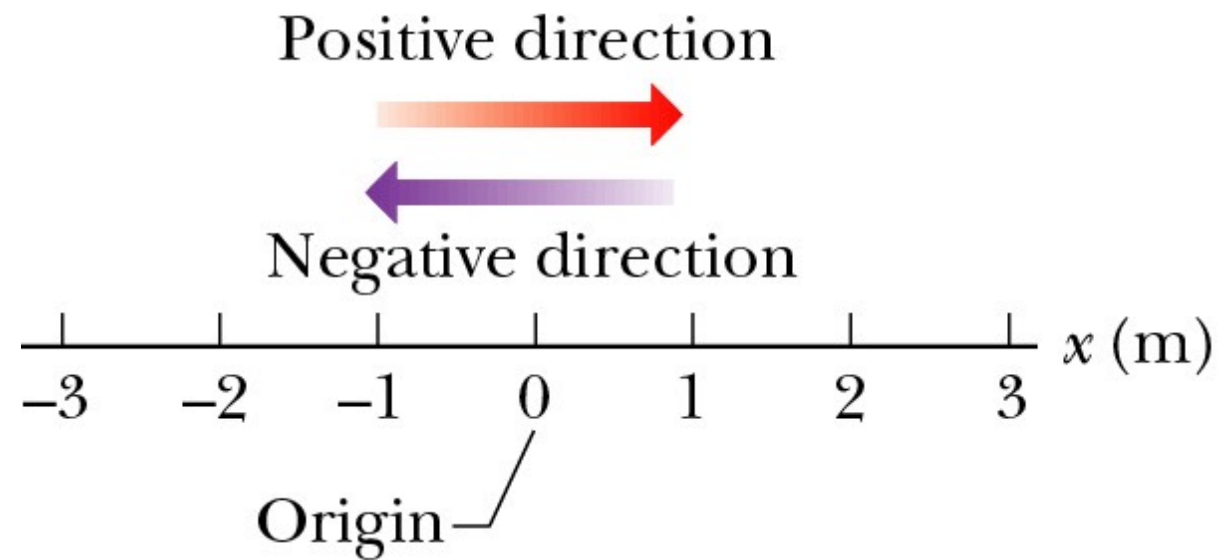
POSITION – WHERE IS AN OBJECT LOCATED?



Reference frames

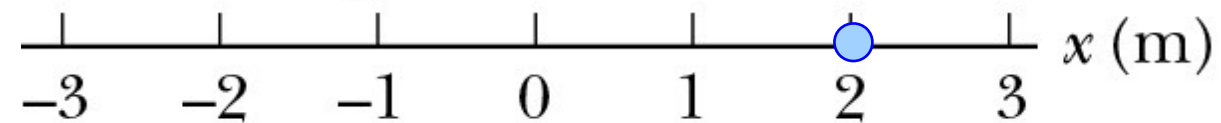
To locate an object means to find its position relative to some **reference point**, often the origin (or zero point).

POSITION – WHERE IS AN OBJECT LOCATED?



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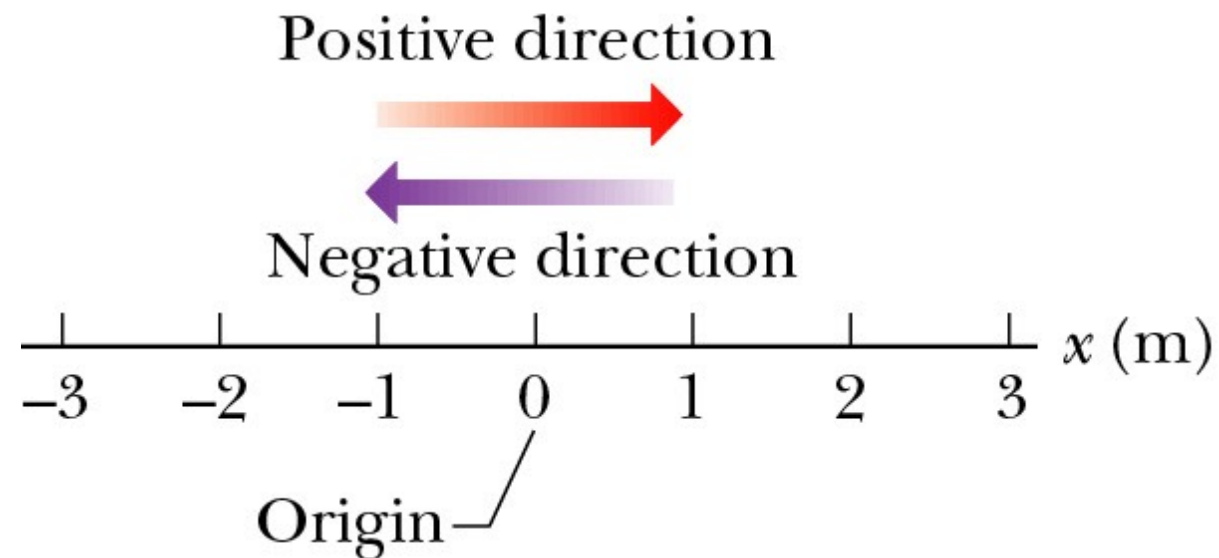


The position of the ball is

$$x = 2\text{m}$$

The + indicates the direction to the right of the origin.

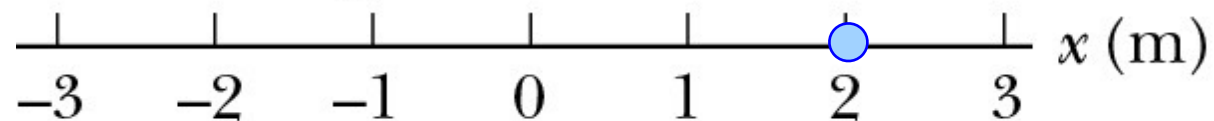
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Reference frames

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The position of the ball is



$$x = 2m$$

The + indicates the direction to the right of the origin.

The position of the ball is

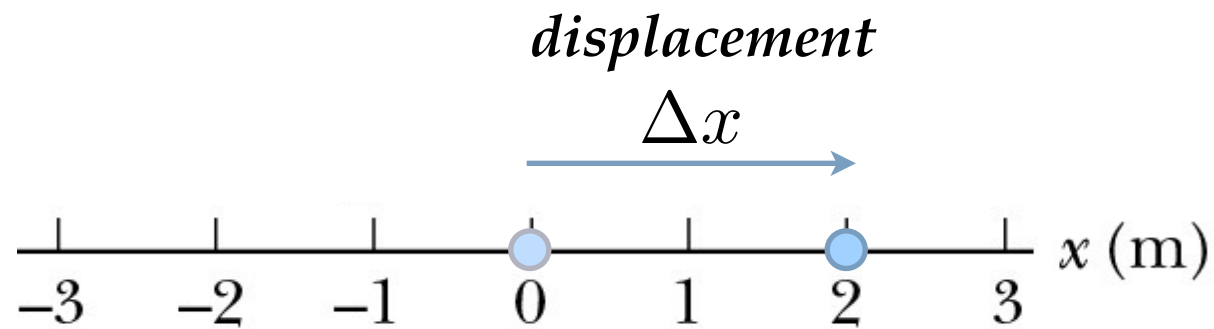


$$x = -2m$$

The - indicates the direction to the left of the origin.

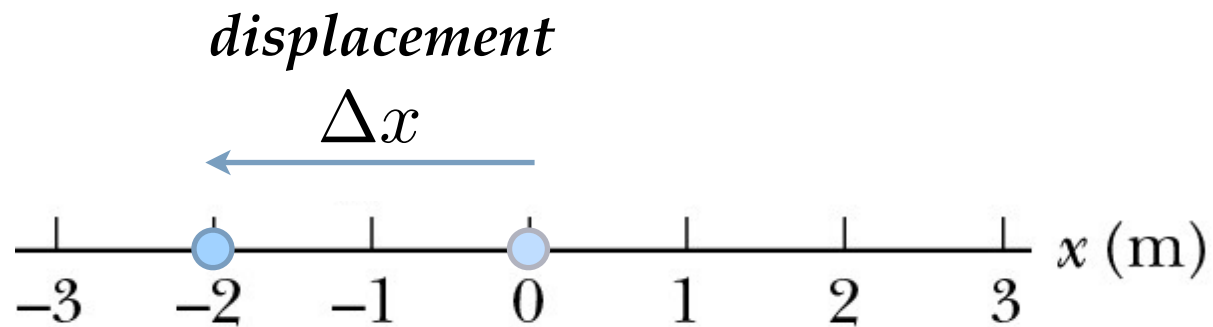
“DISPLACEMENT” & “DISTANCE”

- ‘distance’ = total ground covered while traveling
- ‘displacement’ = *vector* from where you started to where you end up
- displacement and distance can be quite different



$$\begin{aligned}\Delta x &= x_f - x_i \\ &= 2m - 0m \\ &= 2m \quad \text{sign indicates the direction}\end{aligned}$$

& distance = 2m



$$\begin{aligned}\Delta x &= x_f - x_i \\ &= -2m - 0m \\ &= -2m\end{aligned}$$

but distance = 2m

Question (3min)

A car moves 72 km to the right and then reverses direction and travels 120 km to the left.

1. What is the displacement of the car?

- a. 72 km
- b. -72 km
- c. 192 km
- d. -192 km
- e. 48 km
- f. -48 km



2. What is the distance travelled by the car?

- a. 72 km
- b. -72 km
- c. 192 km
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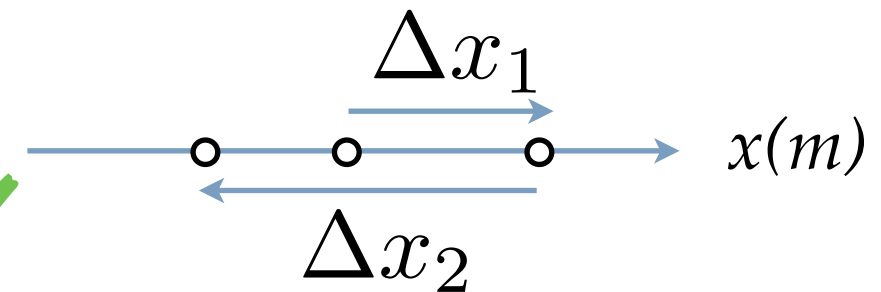
Question (3min)

A car moves 72 km to the right and then reverses direction and travels 120 km to the left.

1. What is the displacement of the car?

$$\begin{aligned}\Delta x &= x_f - x_i \\ &= \Delta x_1 + \Delta x_2 \\ &= (72\text{km} - 0\text{km}) + (-120\text{km}) \\ &= -48\text{km}\end{aligned}$$

- a. 72 km
- b. -72 km
- c. 192 km
- d. -192 km
- e. 48 km
- f. -48 km ✓



2. What is the distance travelled by the car?

$$\begin{aligned}d &= |\Delta x_1| + |\Delta x_2| \\ &= |72\text{km} - 0\text{km}| + |-120\text{km}| \\ &= 192\text{km}\end{aligned}$$

- a. 72 km
- b. -72 km
- c. 192 km ✓
- d. -192 km
- e. 48 km
- f. -48 km

Position - Where you are and **how far** you travelled

Speed – **How fast** you are moving

$$\text{Average speed} = \frac{\text{total distance travelled}}{\text{time lapsed}}$$

Velocity has magnitude and direction – It's a vector!

Examples:

- 30 mi/hr East
- 2 m/s up
- 25 km/s toward Richmond

Speed can be the magnitude of the velocity. It is a scalar.

Average velocity is the displacement divided by the time taken

$$\bar{v} = \frac{\text{displacement}}{\text{time elapsed}} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Dimensions of velocity – [m/s] in SI units (the same as speed!)

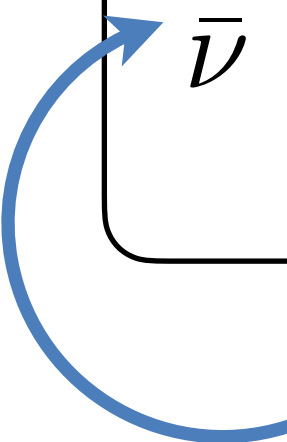
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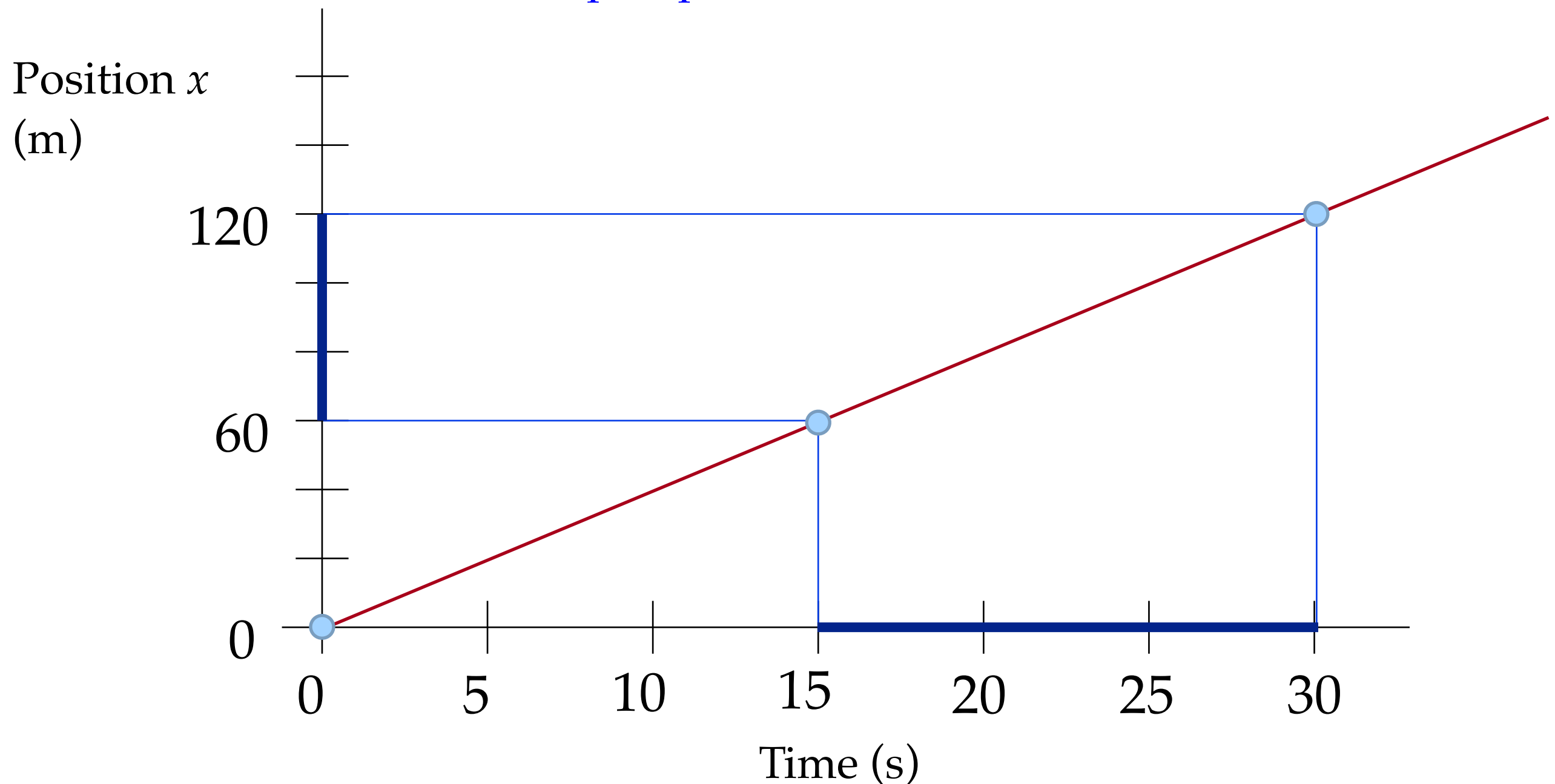

$$\bar{v} = \frac{\text{displacement}}{\text{time elapsed}} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

the little bar on top means “average”

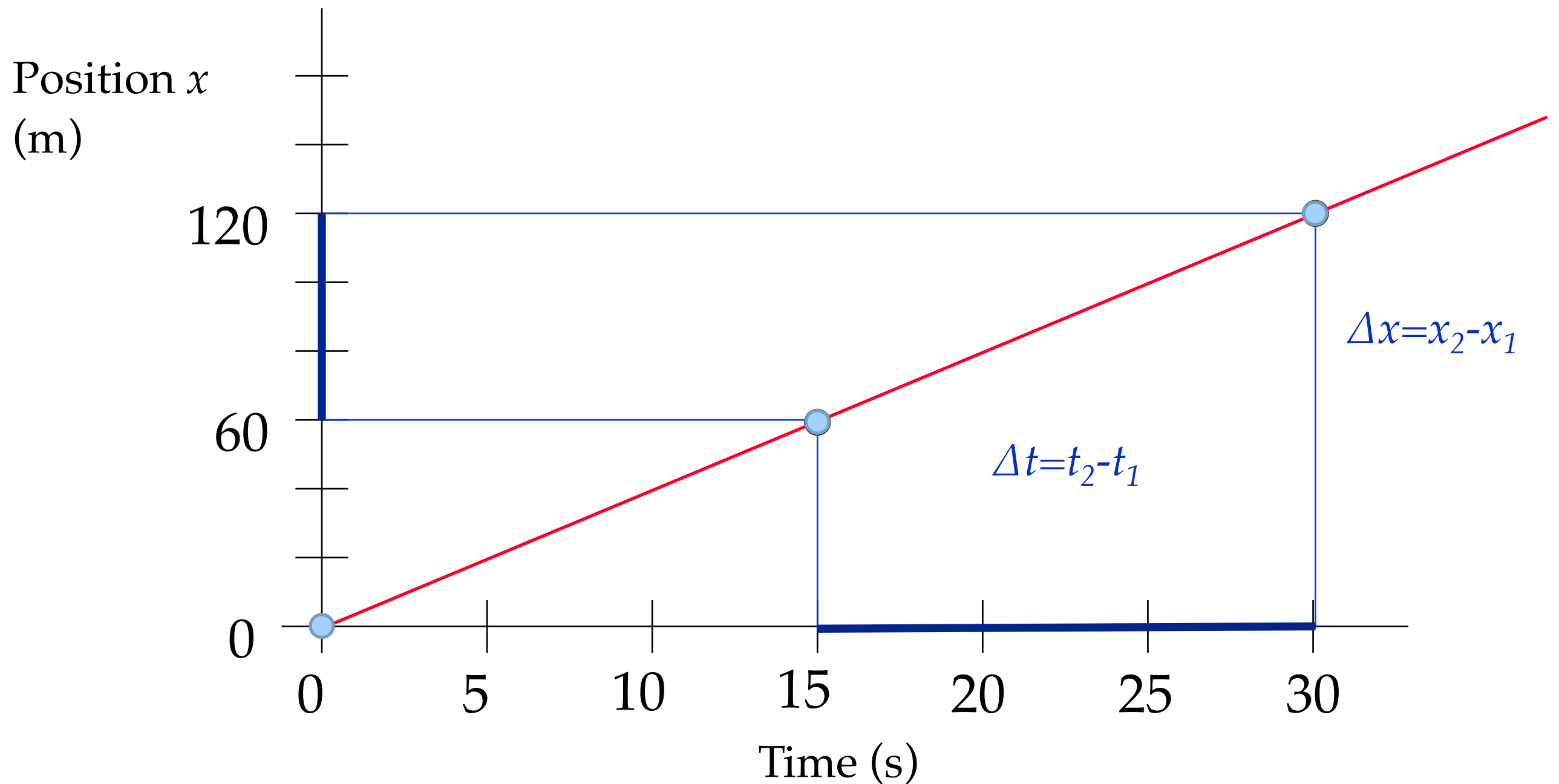
GRAPHING AVERAGE VELOCITY

Imagine that at some point $t_1=15$ s your position is $x_1=60$ m and at $t_2=30$ s your position is $x_2=120$ m.

Let's plot position vs. time:



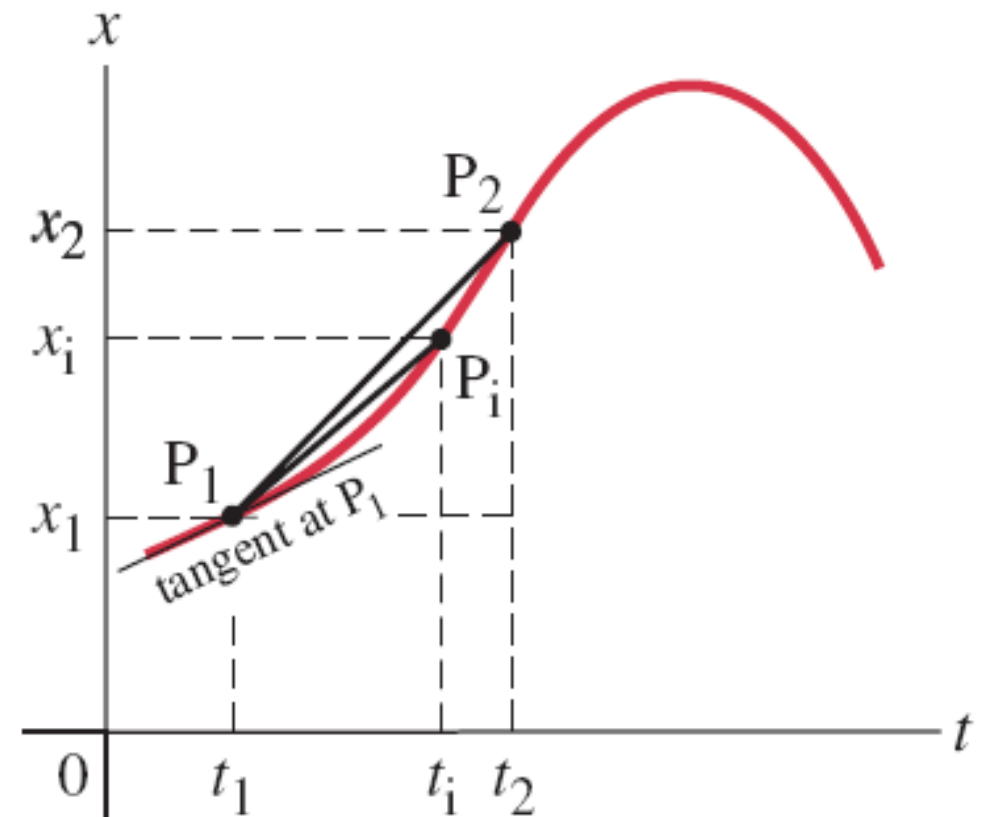
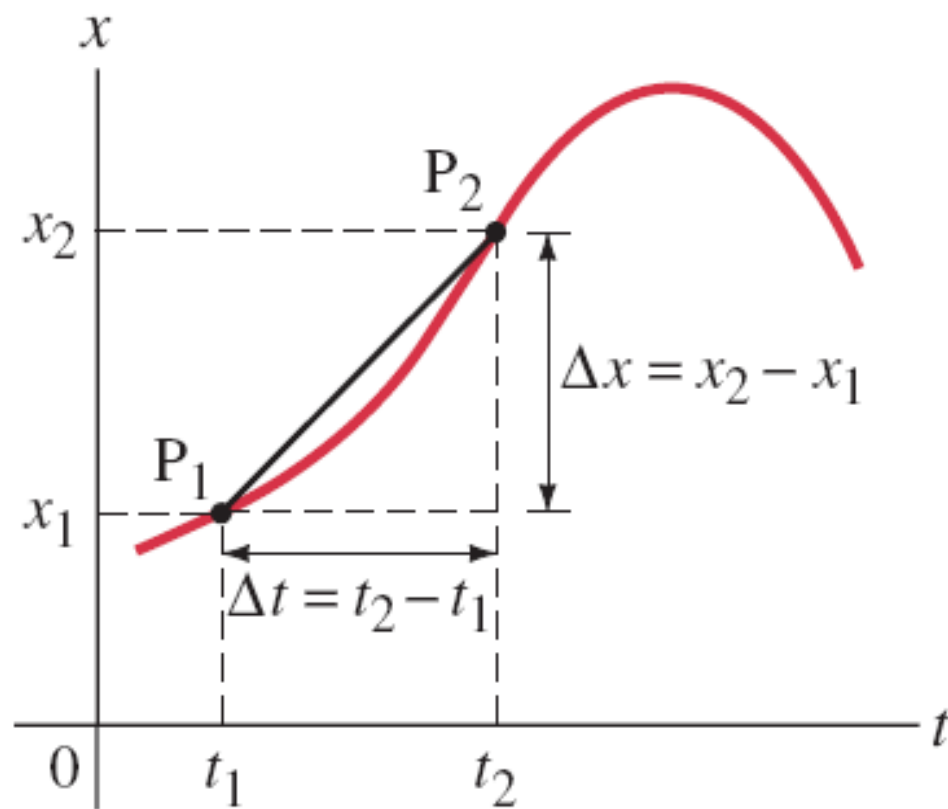
GRAPHING AVERAGE VELOCITY - SLOPE OF THE CURVE



Slope gives velocity: $\bar{v} = \frac{x_2 - x_1}{t_2 - t_1} \Rightarrow \bar{v} = \frac{60\text{m}}{15\text{s}} = 4\frac{\text{m}}{\text{s}}$

AVERAGE VS. INSTANTANEOUS VELOCITY

Instantaneous velocity is a vector defined as “how fast” a particle is moving at a given instant.



$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1}$$

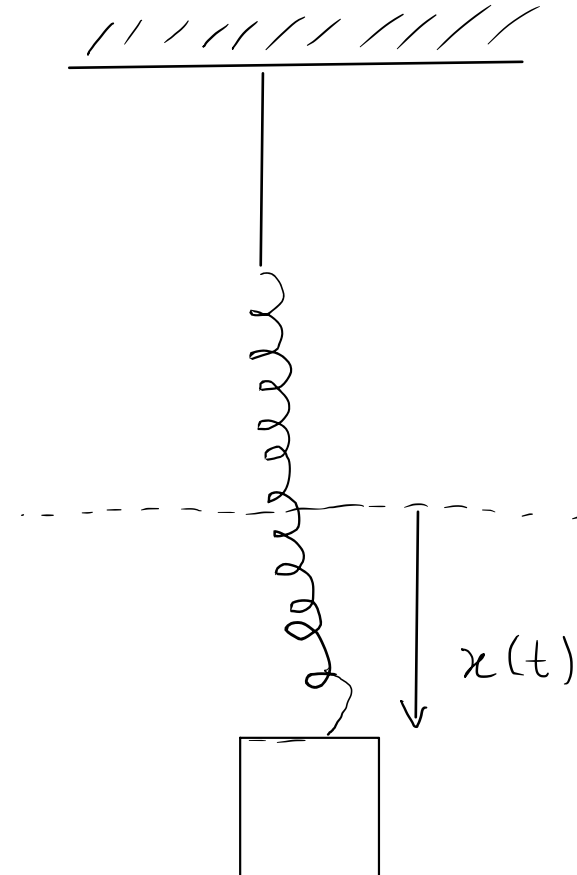
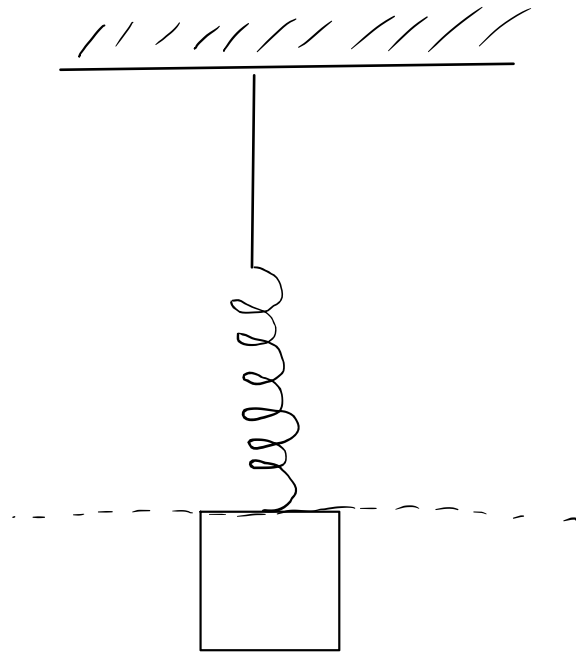


$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Instantaneous velocity is the tangent to the curve at any point.

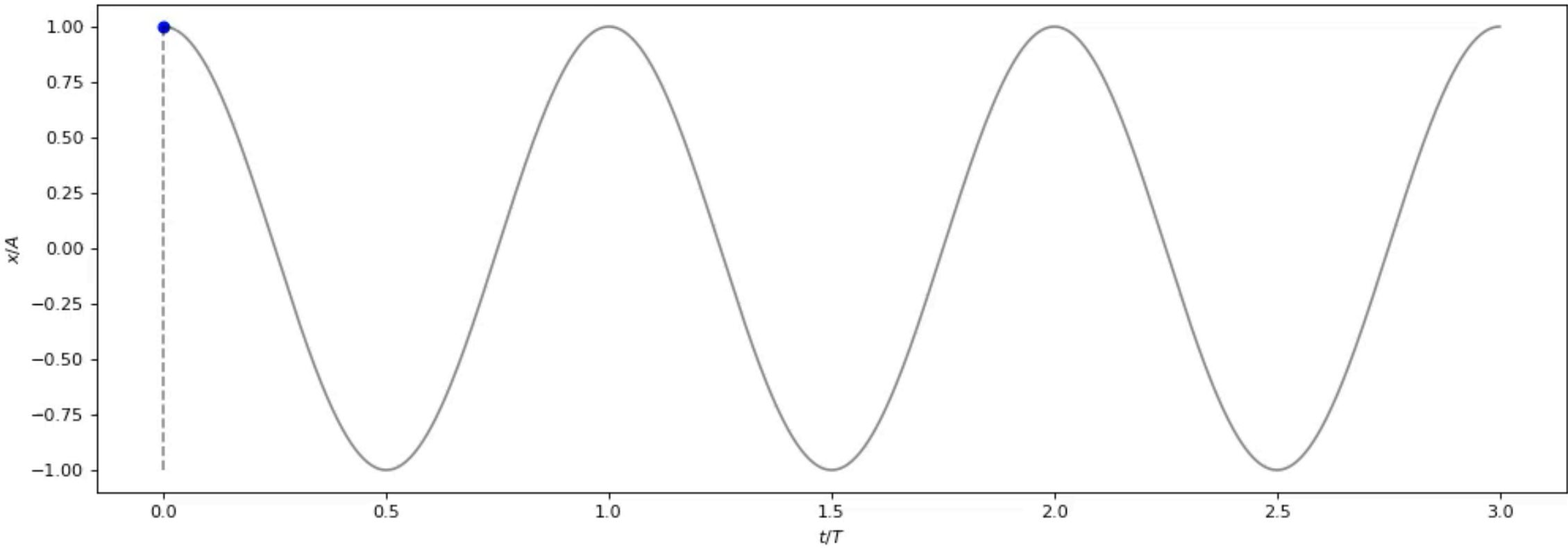
HARMONIC OSCILLATOR

Consider a mass attached to a spring

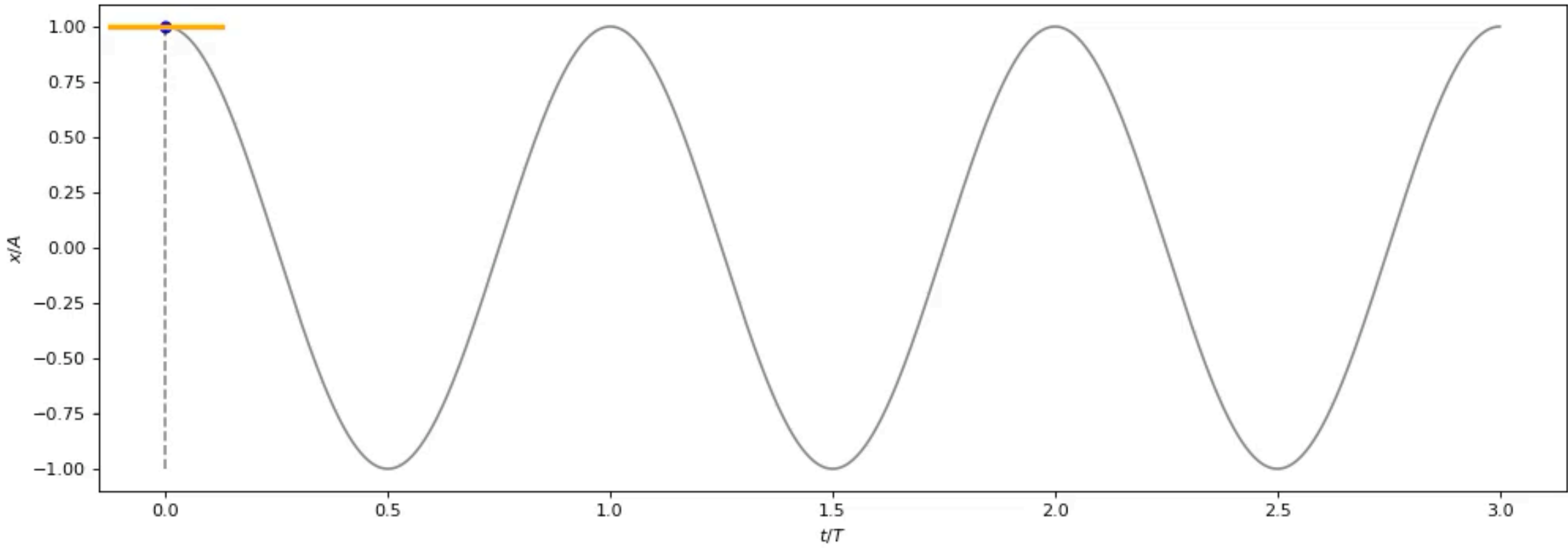


$$x(t) = A \cos\left(\frac{2\pi}{T}t\right)$$

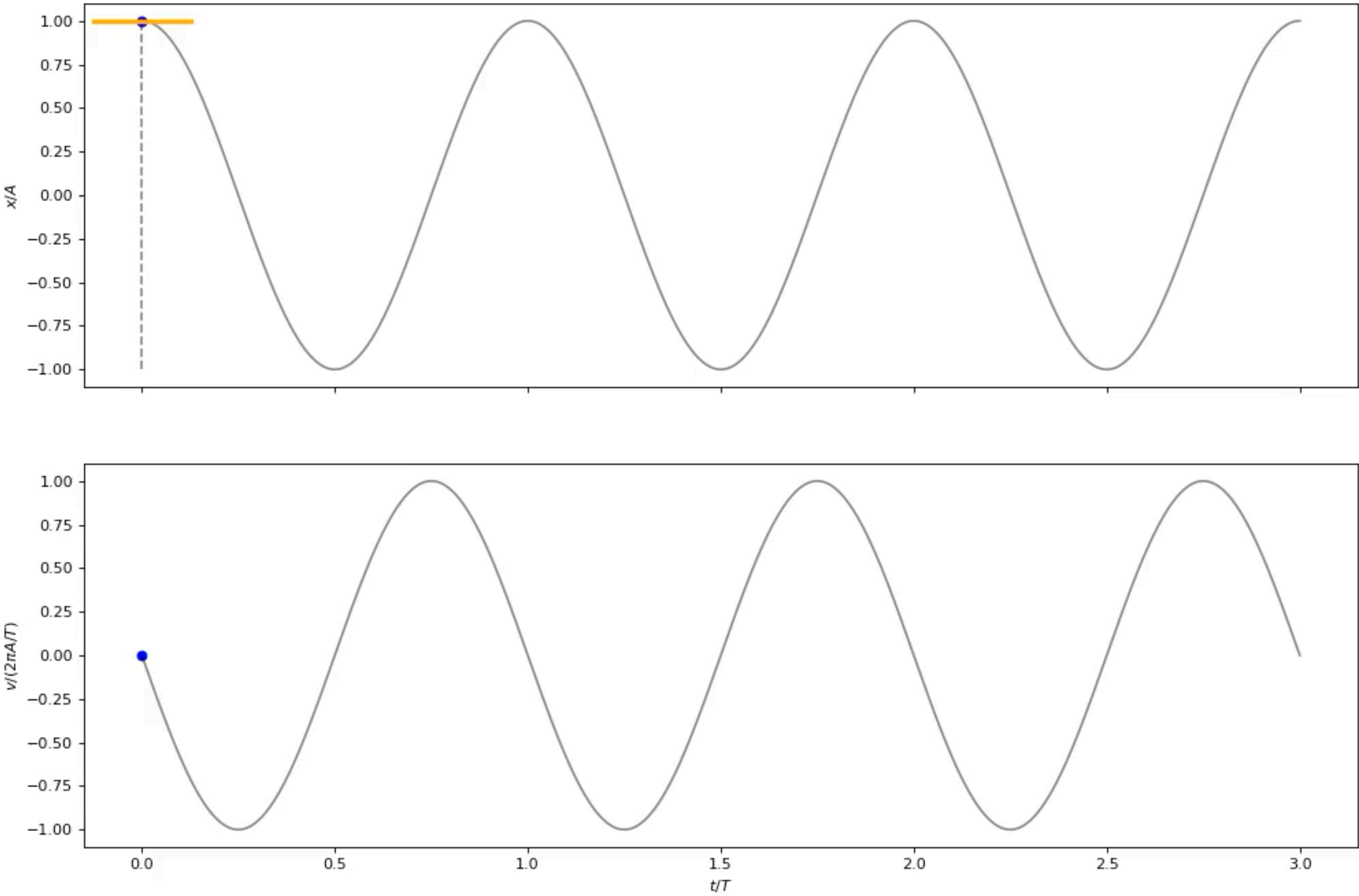
HARMONIC OSCILLATOR POSITION



HARMONIC OSCILLATOR SLOPE



HARMONIC OSCILLATOR VELOCITY



ACCELERATION

Acceleration is the rate of change of velocity

Average acceleration is a vector given as:

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$$

Unit of acceleration: $\left[\frac{m}{s^2} \right]$

ACCELERATION

Acceleration is the rate of change of velocity

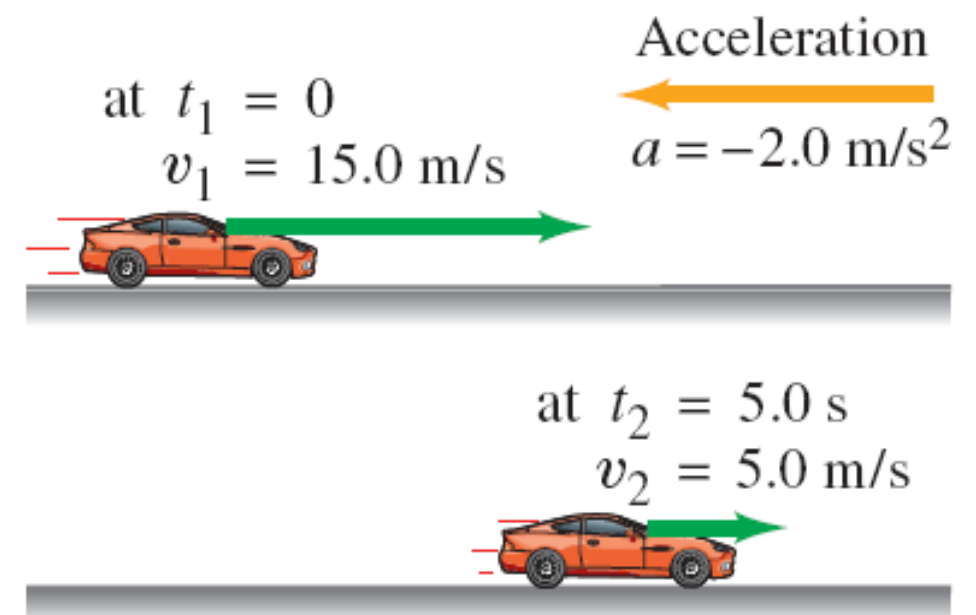
Average acceleration is a vector given as:

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \quad \text{Unit of acceleration: } \left[\frac{m}{s^2} \right]$$

Example: A car moving at initial velocity of 15 m/s slows down to 5 m/s in 5 s. What is the car's average acceleration?

$$\bar{a} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{5 \frac{m}{s} - 15 \frac{m}{s}}{5s - 0s} = -2 \frac{m}{s^2}$$

Velocity is decreasing → acceleration is negative
→ the object is decelerating



CONCEPTUAL QUESTION

If the acceleration is zero, does it mean that the velocity is zero?

- A. Yes.
- B. No. 
- C. Maybe.

Acceleration is the rate of change of velocity.

When acceleration is zero $\rightarrow$ change of velocity is zero $\rightarrow$ velocity is constant (it can but it doesn't have to be zero)

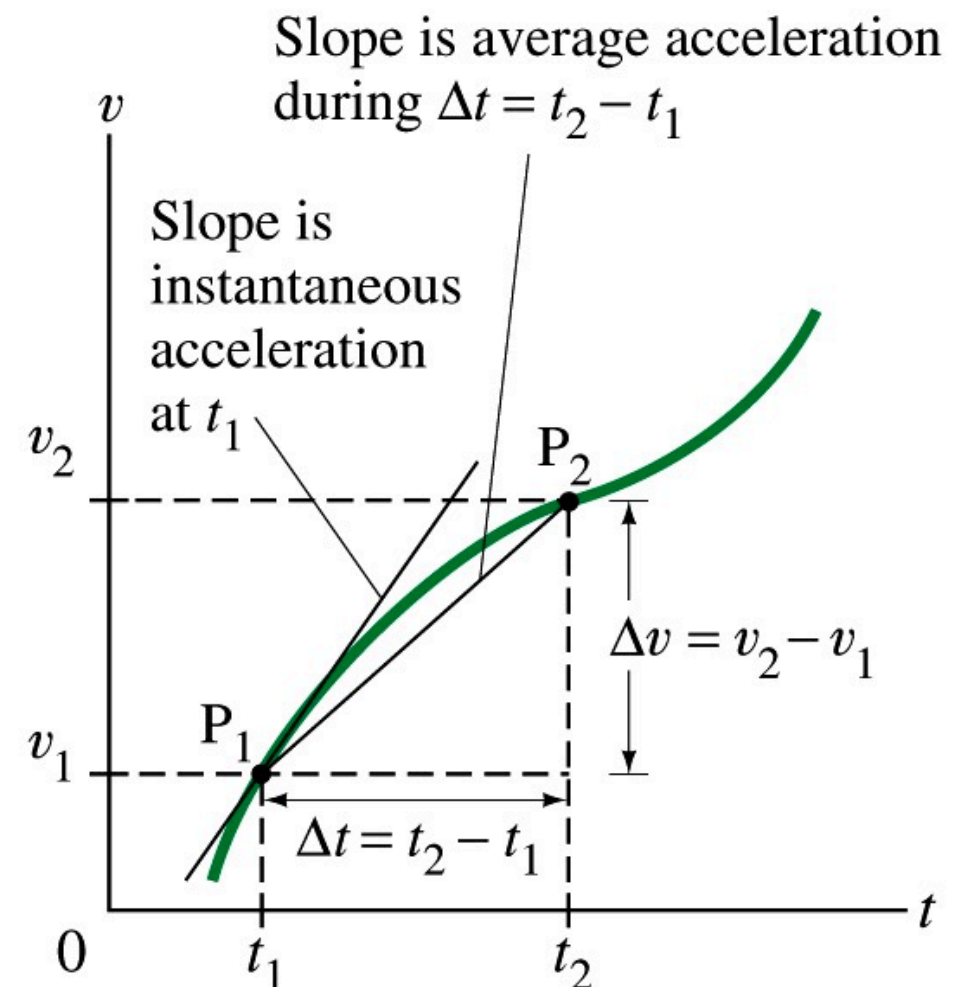
INSTANTANEOUS ACCELERATION

Instantaneous acceleration is the average acceleration in the limit as the time interval becomes infinitesimally short.

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

Unit of acceleration: $\left[\frac{m}{s^2} \right]$

Instantaneous acceleration is the slope of the tangent to the v vs. t curve at that time.



DERIVATIVES

Differential calculus focuses on making sense and defining this limit:

$$\frac{d}{dx}f(x) \equiv \lim_{a \rightarrow 0} \frac{\Delta_a f(x)}{2a} = \lim_{a \rightarrow 0} \frac{f(x+a) - f(x-a)}{2a}$$

derivative of $f(x)$

This is of course the instantaneous rate of change of the $f(x)$.

Determining this for different functions analytically is not always easy...

...numerically it is much easier 🧐

There are alternative derivations as well...

$$\frac{d}{dx}f(x) \equiv \lim_{a \rightarrow 0} \frac{f(x+a) - f(x)}{a}$$

Forward approximation of $\frac{d}{dx}f(x)$

EXERCISE #1 - DEFINE DERIVATIVE

Write code for a “*derivative*” of any function, f , at x for a non-zero value of a .

Note, in Python, you can define functions which take function as inputs. Here is an example of this.

```
def fin_deriv(f, x, a):  
    """  
    this defines the rate of change of the function f  
    as a function of x  
  
    the final point is at  $x_f = x + a$   
    the initial point is at  $x_i = x - a$   
    the difference between these two is  $x_f - x_i = 2*a$   
    """  
  
    "the numerator"  
    num = f(x+a) - f(x-a)  
    "the denominator"  
    denum = 2.0* a  
  
    return num / denum
```

look inside my code 😎

TABLE OF USEFUL DERIVATIVES

A **derivative** is a function the rate of change of that function.

Here is a list of derivatives of some of the most common functions, where **c = constant**:

$$\frac{dc}{dx} = 0$$

$$\frac{d}{dx} \sin(cx) = c \cos(cx)$$

$$\frac{d}{dx} \ln(cx) = \frac{1}{x}$$

$$\frac{d}{dx} (cx^n) = cnx^{n-1}$$

$$\frac{d}{dx} \cos(cx) = -c \sin(cx)$$

$$\frac{d}{dx} e^{cx} = ce^{cx}$$

if not yet, you should
memorize them 🧐

TESTING CONVERGENCE - PART1

Remember, the derivative is defined in the limit that $a = 0$. So the first thing we should do is test the “*convergence*” of our code. In particular, we want to plot the resulting function at some value of x as a function of a .

WARNING: In practice, we cannot take a to be exactly zero. This is not well defined numerically. Instead, we need to verify that if a is sufficiently small, our answer no longer depends on the value chosen for a . If we reach a value of a , and our answer no longer varies we would say that the answer has converged.

We can visualize the answer converging to a desired answer, but we would also like to introduce a quantity that measures the error of our result. If the derivative is independent of a , then the derivative of this with respect to a must be zero.

To define this, let us follow these steps.

1) Calculate the derivative first at $a + \delta$: $\frac{\Delta_{a+\delta} f(x)}{2(a + \delta)}$

2) Then you calculate at a smaller value, a : $\frac{\Delta_a f(x)}{2a}$

TESTING CONVERGENCE - PART2

3) Take the difference between these two values and divide by δ .

4) In order to see this better, we will multiply this derivative by 100:

$$\sigma_a = \frac{100}{\delta} \times \left| \frac{\Delta_a f(x)}{2a} - \frac{\Delta_{a+\delta} f(x)}{2(a+\delta)} \right|$$

EXERCISE #2 - TEST CONVERGENCE

In what follows we will vary $a = [10^{-5}, .1)$ and fix $x = 1$.

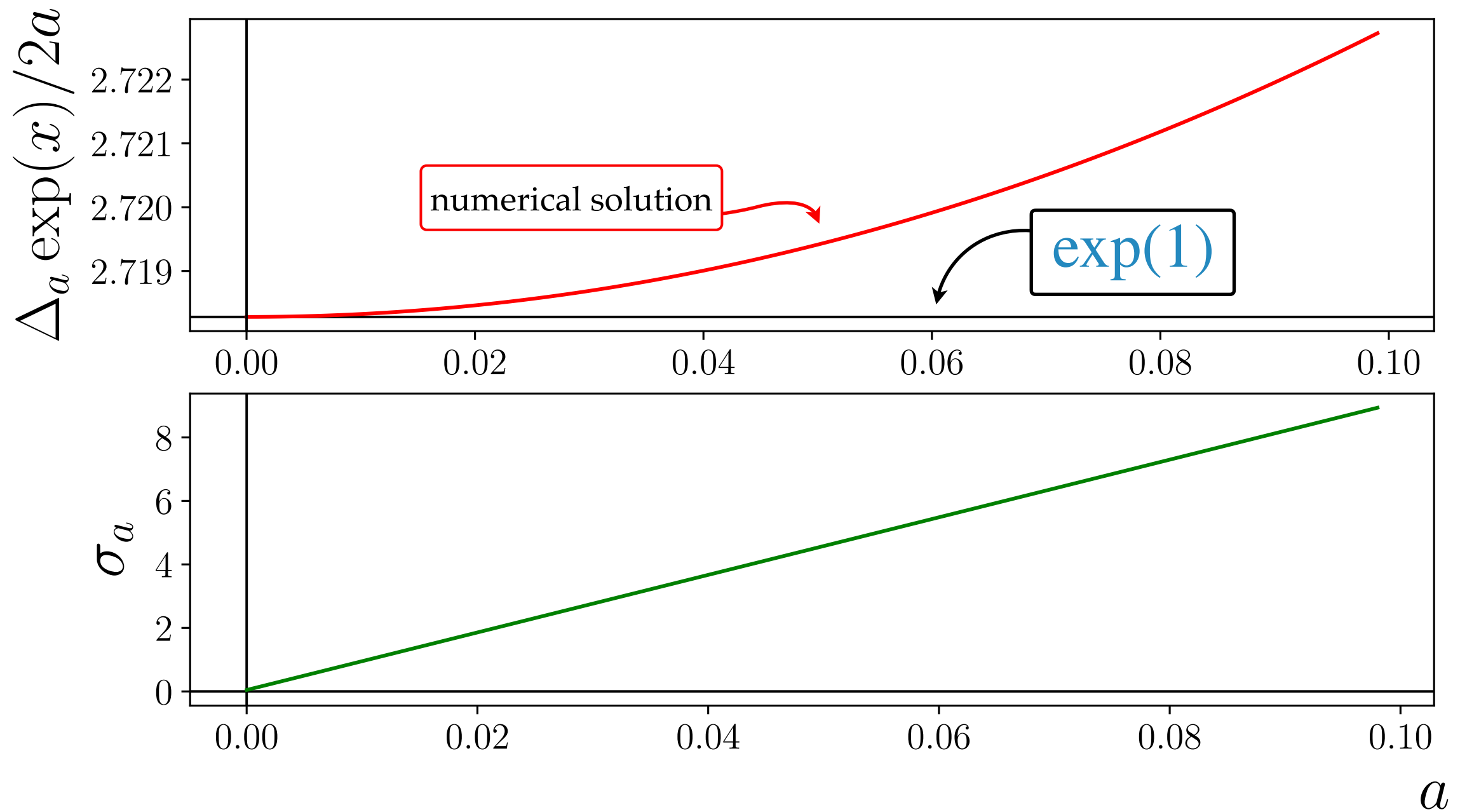
```
x = 1  
a=np.arange(pow(10,-5), .1, pow(10,-3))
```

Note, I chose to vary a in steps of 10^{-3} , but you are welcomed to try a different step size.

Let us test your code by taking the numerical derivative of the following functions:

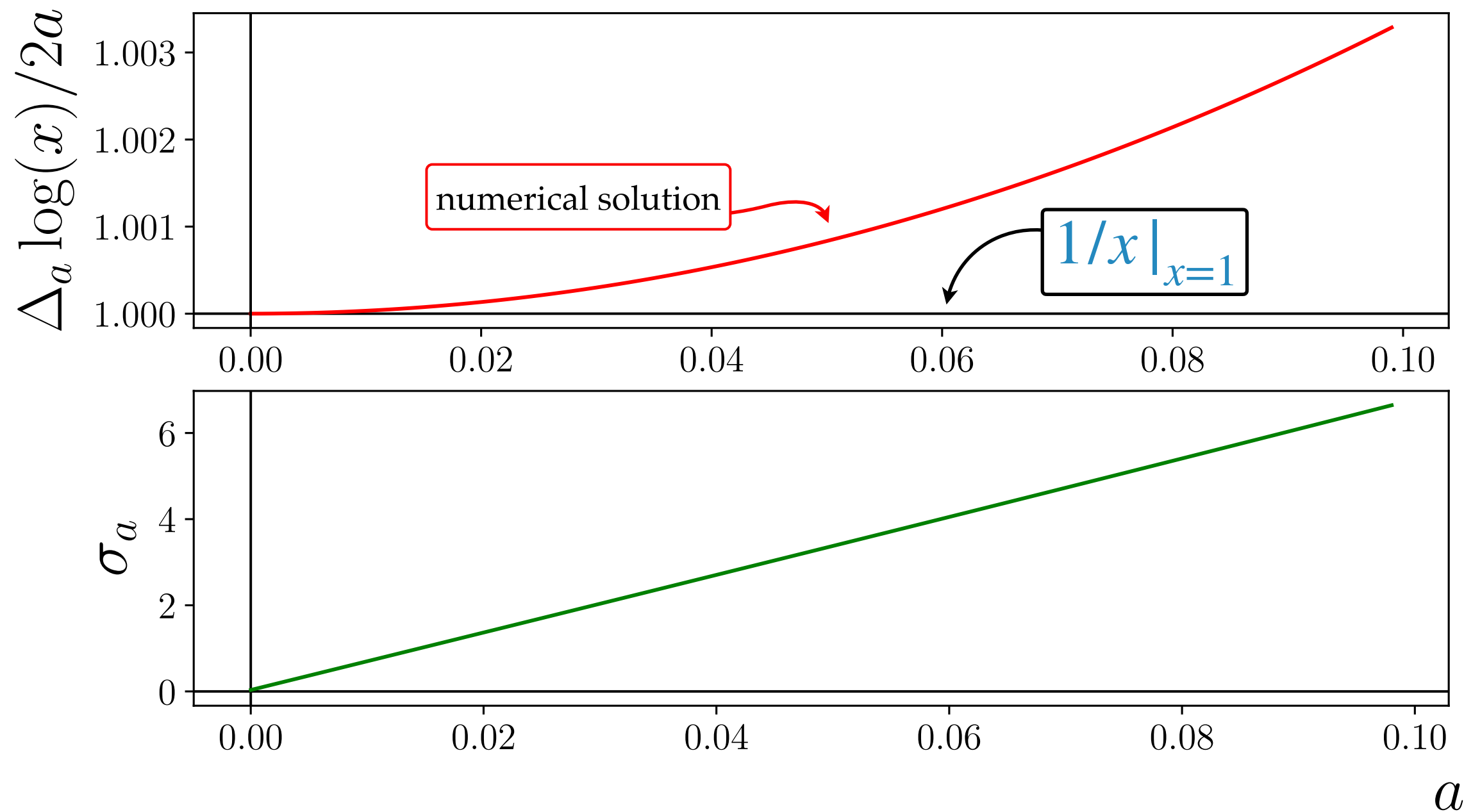
$\exp(x)$, $\log(x)$, $\sin(x)$, $\cos(x)$, $\frac{x^3}{6} + \frac{x^4}{8}$

EXERCISE #2 - TEST CONVERGENCE: $\exp(x)$



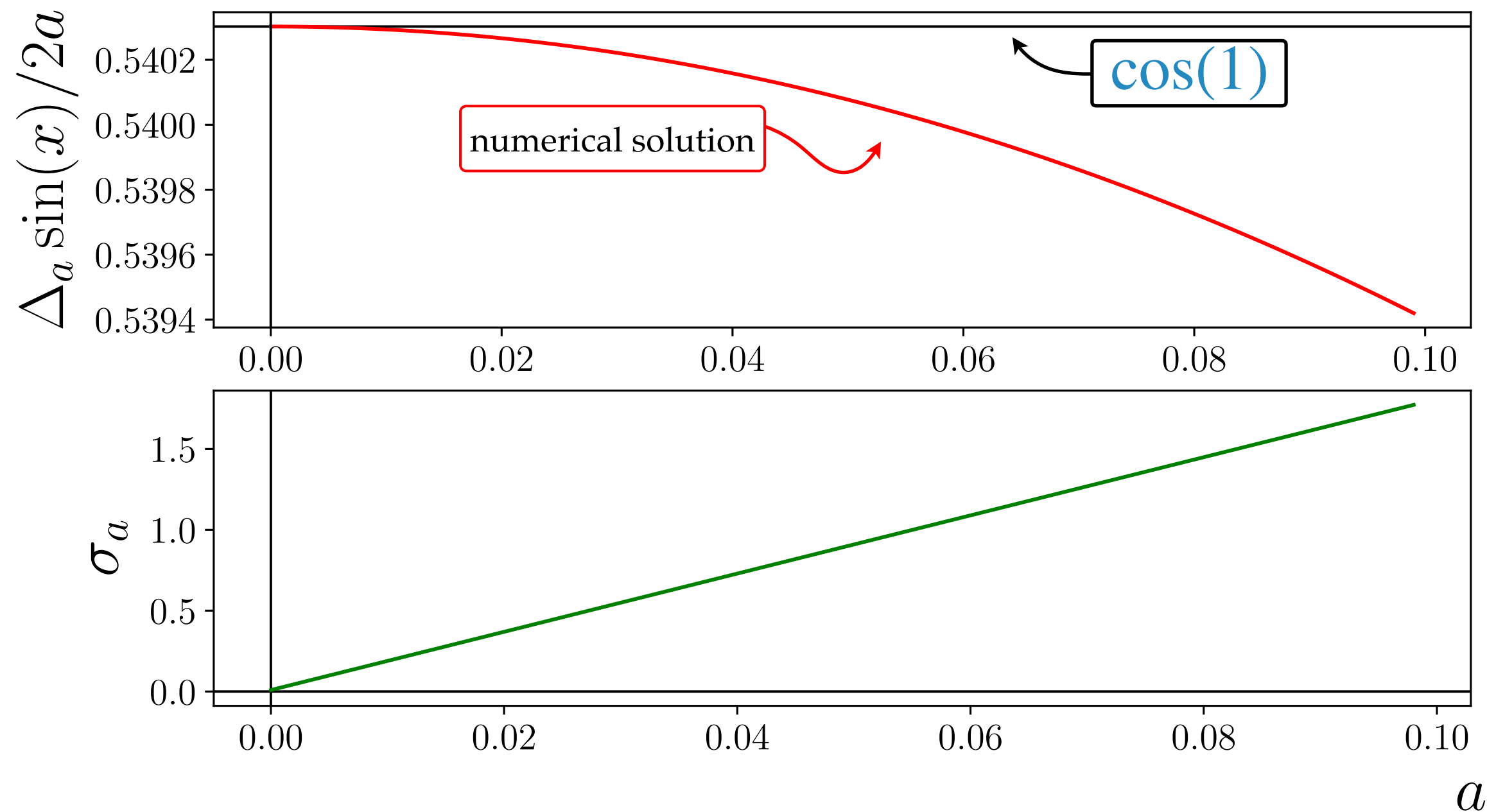
$$\sigma_a = \frac{100}{\delta} \times \left| \frac{\Delta_a f(x)}{2a} - \frac{\Delta_{a+\delta} f(x)}{2(a+\delta)} \right|$$

EXERCISE #2 - TEST CONVERGENCE: $\log(x)$



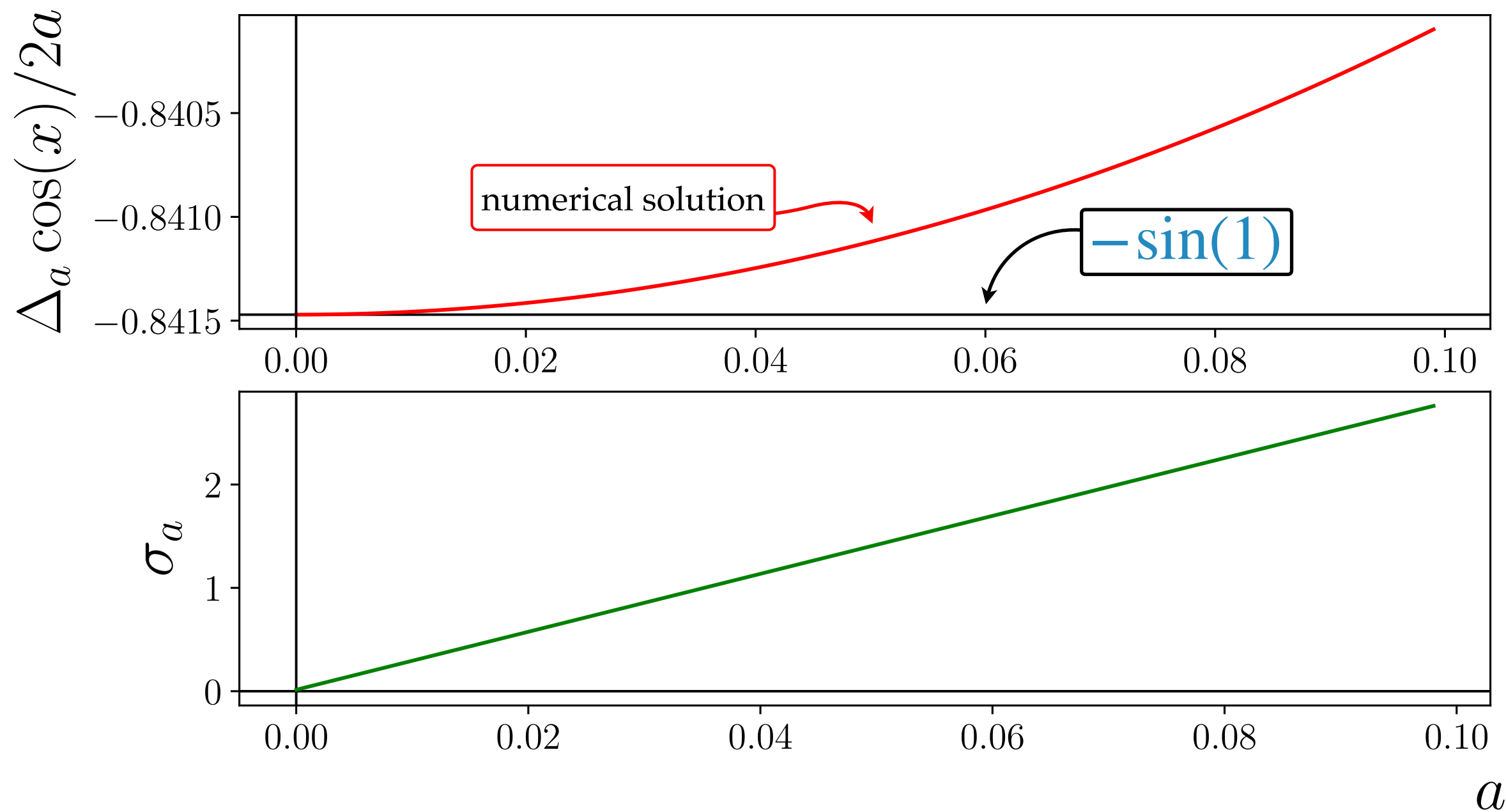
$$\sigma_a = \frac{100}{\delta} \times \left| \frac{\Delta_a f(x)}{2a} - \frac{\Delta_{a+\delta} f(x)}{2(a+\delta)} \right|$$

EXERCISE #2 - TEST CONVERGENCE: $\sin(x)$



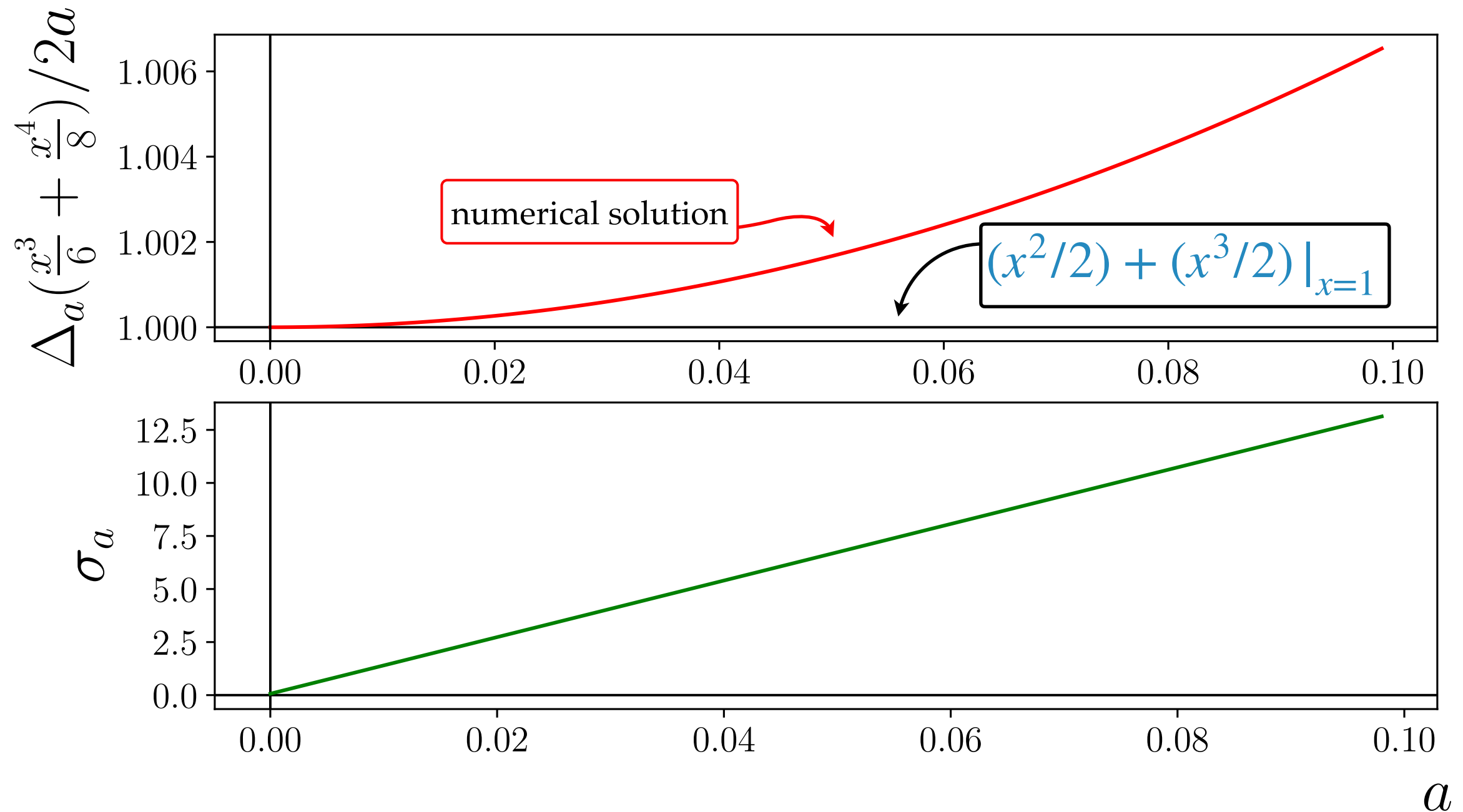
$$\sigma_a = \frac{100}{\delta} \times \left| \frac{\Delta_a f(x)}{2a} - \frac{\Delta_{a+\delta} f(x)}{2(a+\delta)} \right|$$

EXERCISE #2 - TEST CONVERGENCE: $\cos(x)$



$$\sigma_a = \frac{100}{\delta} \times \left| \frac{\Delta_a f(x)}{2a} - \frac{\Delta_{a+\delta} f(x)}{2(a+\delta)} \right|$$

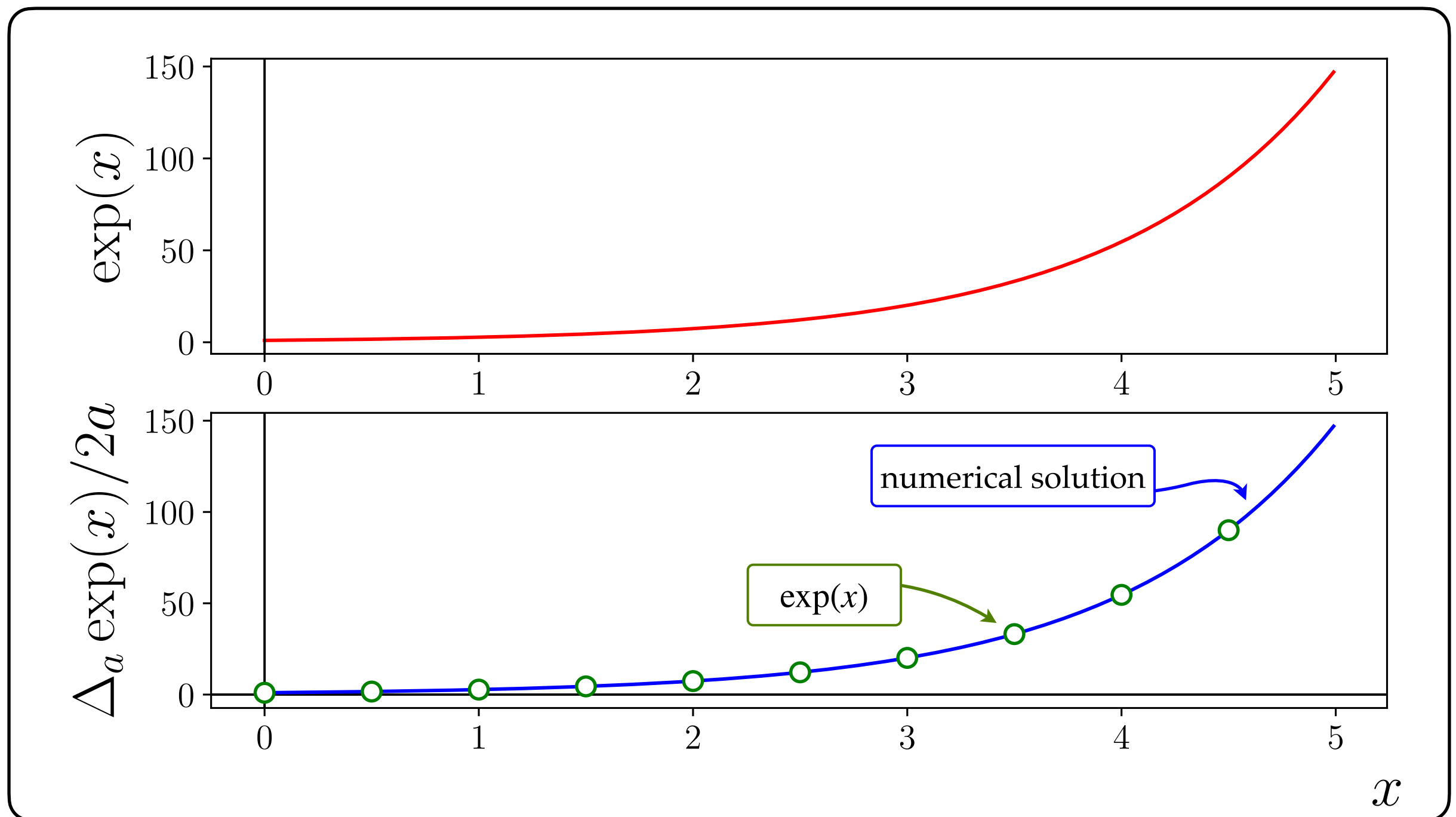
EXERCISE #2 - TEST CONVERGENCE: $(x^3/6) + (x^4/8)$



$$\sigma_a = \frac{100}{\delta} \times \left| \frac{\Delta_a f(x)}{2a} - \frac{\Delta_{a+\delta} f(x)}{2(a+\delta)} \right|$$

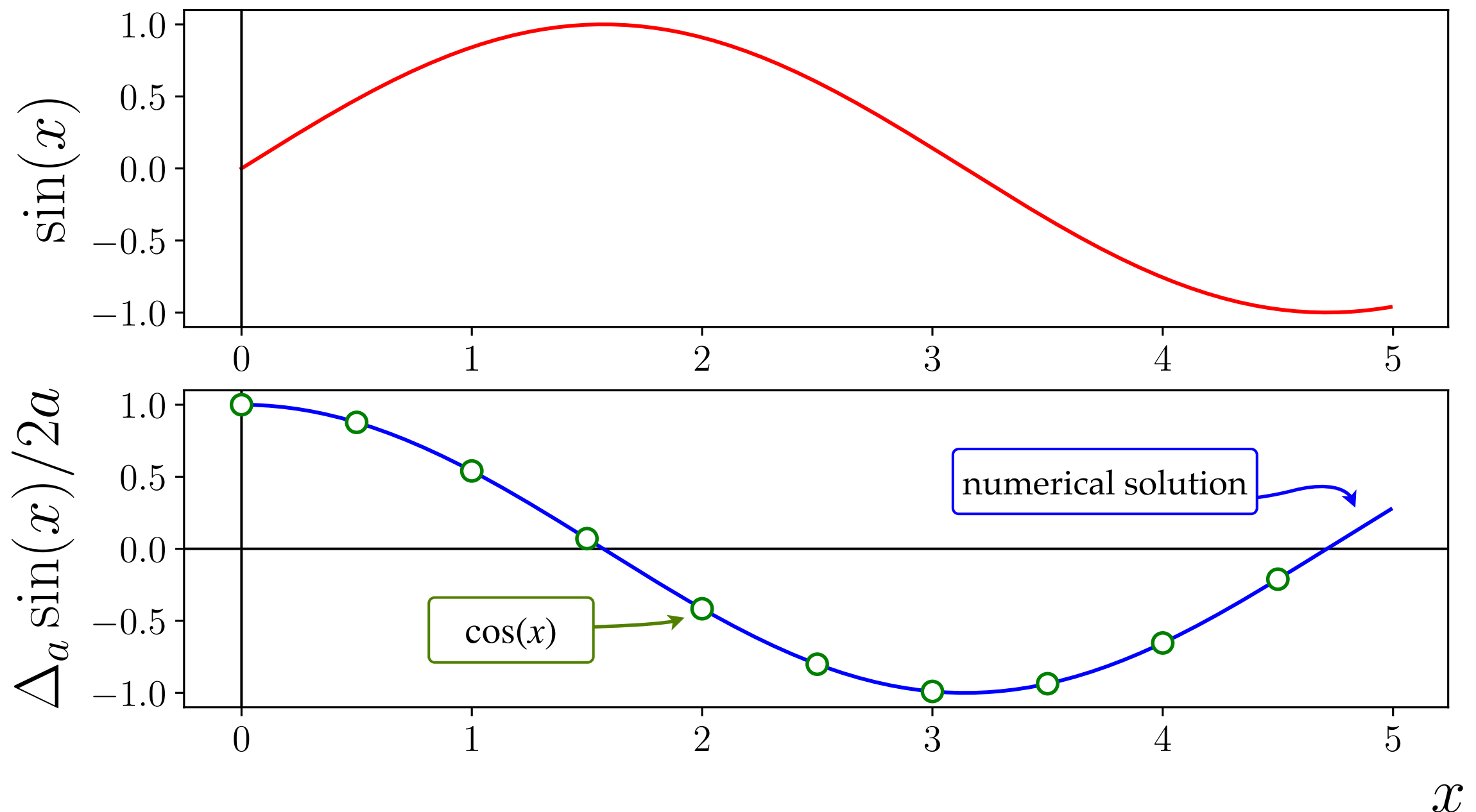
EXERCISE #3 - PLOT DERIVATIVES AS A FUNCTION OF X

Having tested the convergence of your code, you are now in a place to plot the derivative of the same functions as a function of x . We will fix $a = 10^{-3}$.



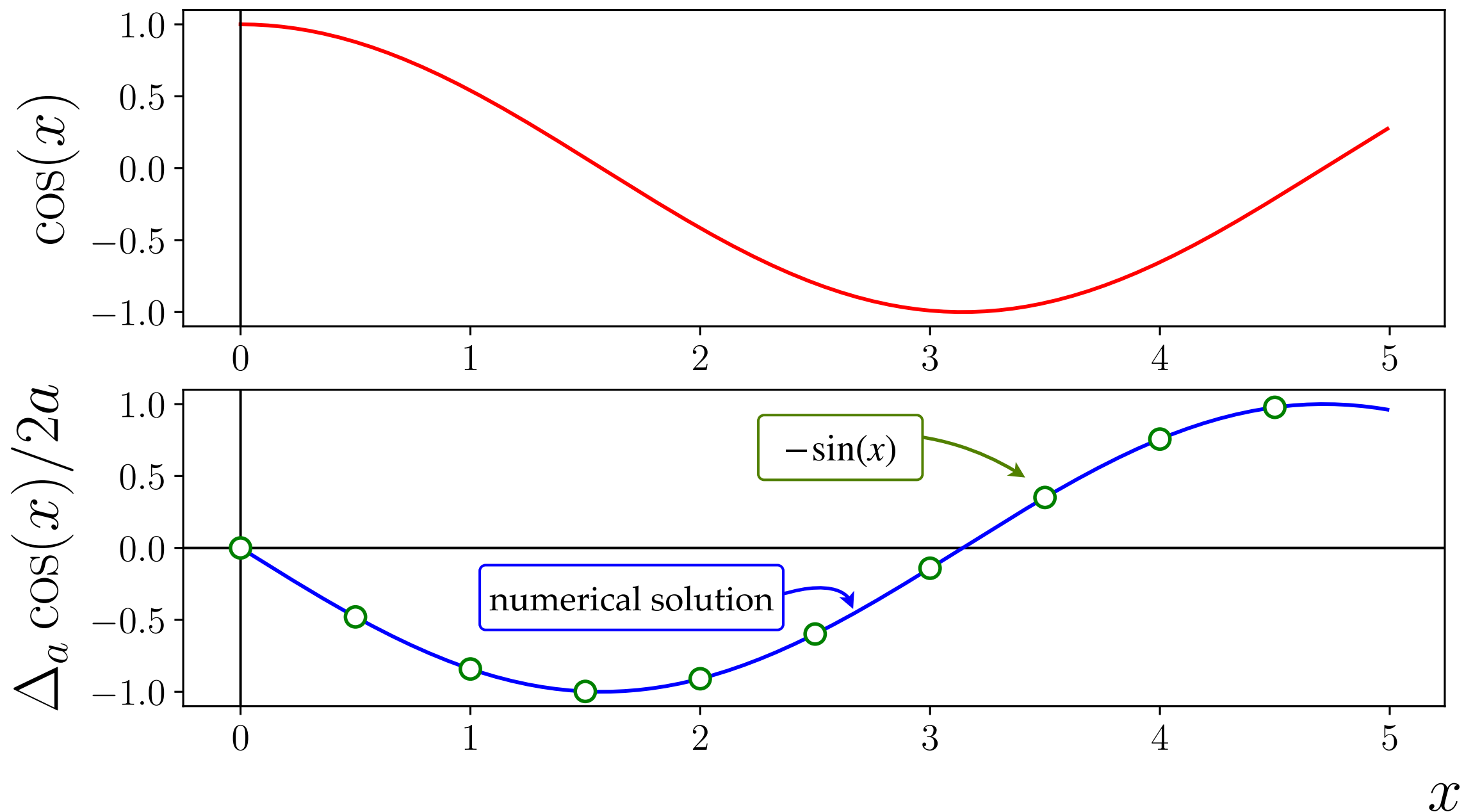
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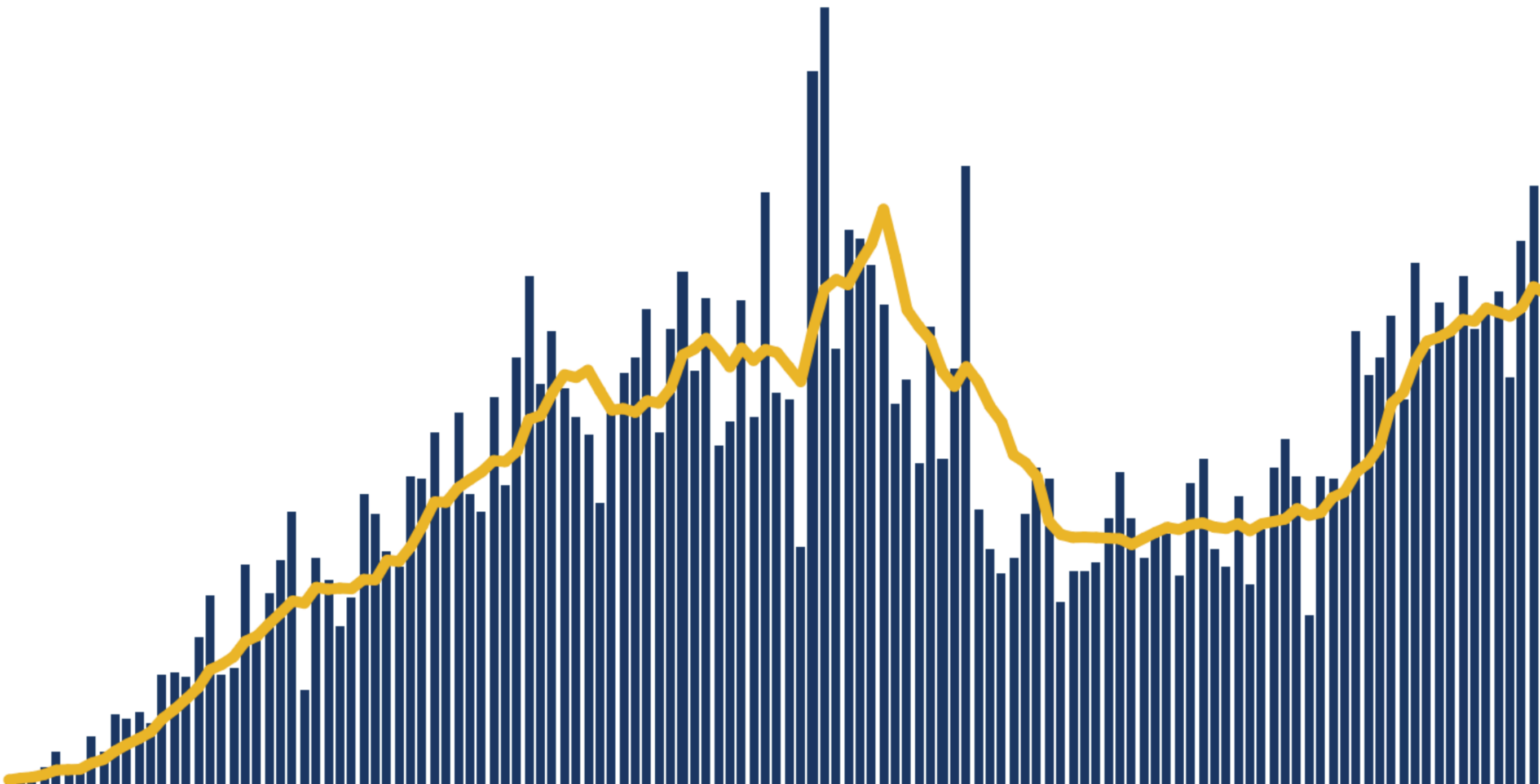


EXERCISE #3 - PLOT DERIVATIVES AS A FUNCTION OF X

Having tested the convergence of your code, you are now in a place to plot the derivative of the same functions as a function of x . We will fix $a = 10^{-3}$.



FIRST ORDER DIFFERENTIAL EQUATION



DIFFERENTIAL EQUATIONS

Differential equations are equations that relate one or more functions and their derivations.

Here is a simple example of a differential equation: $\frac{d}{dx}f(x) = g(x)$.

In general, differential equations can take a more complicated form.

A significant fraction of the most important equations in Physics are differential equations or can be written as such. Examples include:

- $F = ma$, which can be written as $F = m \frac{d^2}{dt^2}x(t)$
- Wave equations
- Schrödinger Equation
- Maxwells Equations

SOLVING NUMERICALLY DIFFERENTIAL EQUATIONS - PART1

Although finding an analytical solution to differential equations may seem like a daunting task, obtaining a numerical solution is more straightforward. In fact, all we need is skills we have developed and a bit of algebra.

To see this, let us consider the simple example previously mentioned: $\frac{d}{dx}f(x) = g(x)$.

Numerically we know we need to replace the derivative with a finite difference by introducing a finite value of a

$$\frac{d}{dx}f(x) = \lim_{a \rightarrow 0} \frac{\Delta_a f(x)}{2a} = \lim_{a \rightarrow 0} \frac{f(x+a) - f(x-a)}{2a}$$

Just as before, we will keep a fixed and check that our results are independent of it. By replacing the derivative with this expression, we are able to rewrite this equation as an algebraic one

$$\frac{f(x+a) - f(x-a)}{2a} = g(x).$$

We can rewrite this by solving for $f(x+a)$

$$f(x+a) = f(x-a) + g(x)2a.$$

SOLVING NUMERICALLY DIFFERENTIAL EQUATIONS - PART2

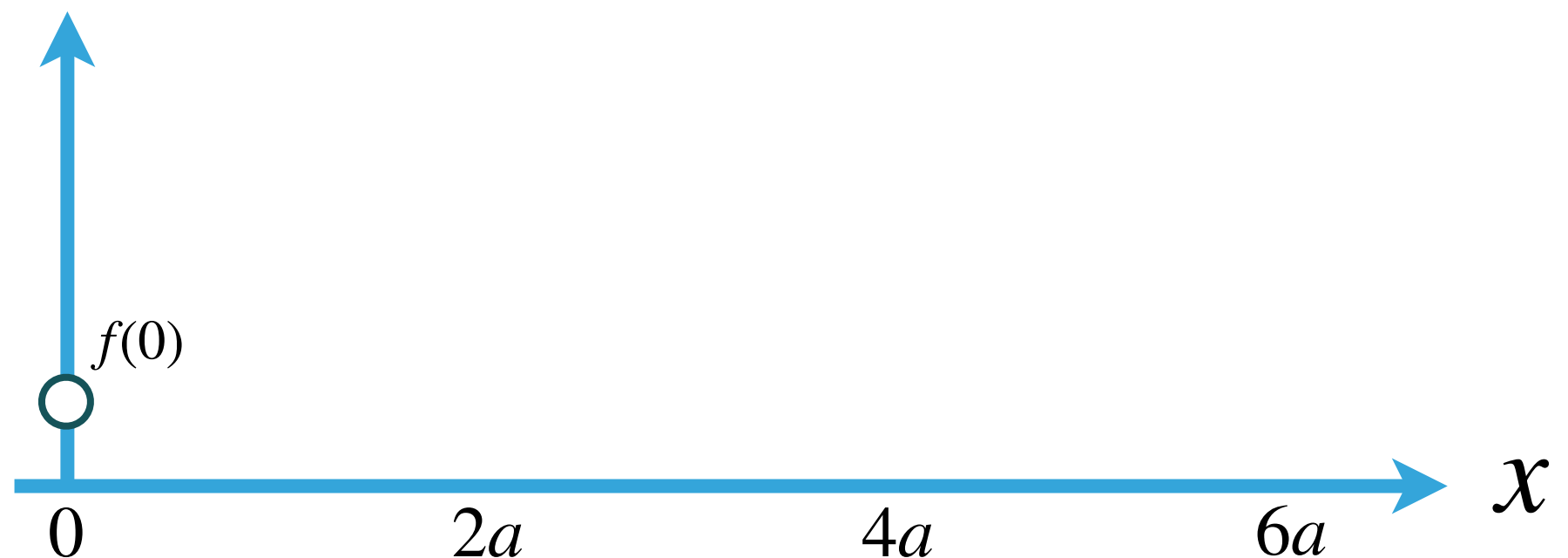
We can rewrite this by replacing $x = x_0 + a$: $f(x_0 + 2a) = f(x_0) + g(x_0 + a)2a$.

Since x_0 is the name of a generic variable, we can just leave it as x

$$f(x + 2a) = f(x) + g(x + a)2a.$$

Notice, if we know the first value of our function, we can determine all the other points using this equation.

Let the first point be a $x = 0$ with value $f(0)$.



SOLVING NUMERICALLY DIFFERENTIAL EQUATIONS - PART2

We can rewrite this by replacing $x = x_0 + a$: $f(x_0 + 2a) = f(x_0) + g(x_0 + a)2a$.

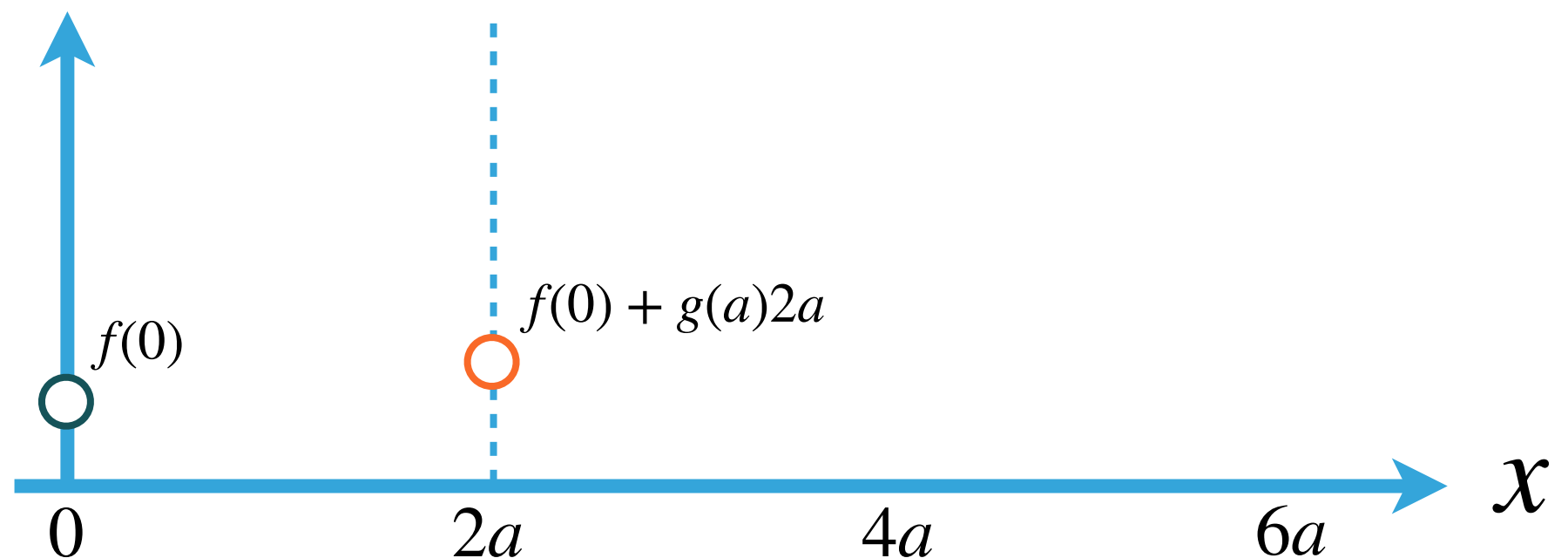
Since x_0 is the name of a generic variable, we can just leave it as x

$$f(x + 2a) = f(x) + g(x + a)2a.$$

Notice, if we know the first value of our function, we can determine all the other points using this equation.

Let the first point be a $x = 0$ with value $f(0)$.

Then, the next point will be at $x = 2a$ with value $f(2a) = f(0) + g(a)2a$.



SOLVING NUMERICALLY DIFFERENTIAL EQUATIONS - PART2

We can rewrite this by replacing $x = x_0 + a$: $f(x_0 + 2a) = f(x_0) + g(x_0 + a)2a$.

Since x_0 is the name of a generic variable, we can just leave it as x

$$f(x + 2a) = f(x) + g(x + a)2a.$$

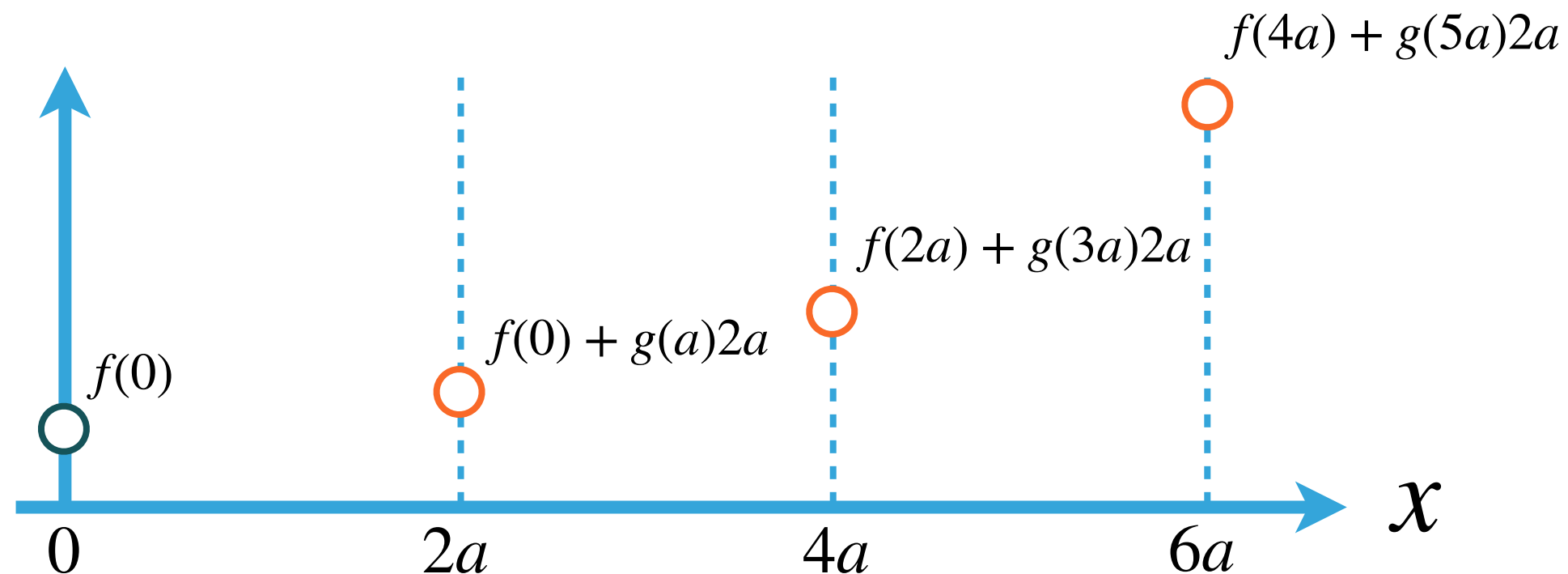
Notice, if we know the first value of our function, we can determine all the other points using this equation.

Let the first point be a $x = 0$ with value $f(0)$.

Then, the next point will be at $x = 2a$ with value $f(2a) = f(0) + g(a)2a$.

Follow by $f(4a) = f(2a) + g(3a)2a = f(0) + g(3a)2a + g(a)2a$

and $f(6a) = f(4a) + g(5a)2a = f(0) + g(5a)2a + g(3a)2a + g(a)2a$.



EXERCISE #4 - WRITE CODE THAT SOLVES DIFF. EQ. FOR ANY $g(x)$

Write code for obtaining $f(x)$ for a range of values of x given any function of $g(x)$ and some initial value of $f(0)$.

To do this, you can write a “for loop” that reiterated the equation we obtained in the previous slide:

$$f(x + 2a) = f(x) + g(x + a)2a.$$

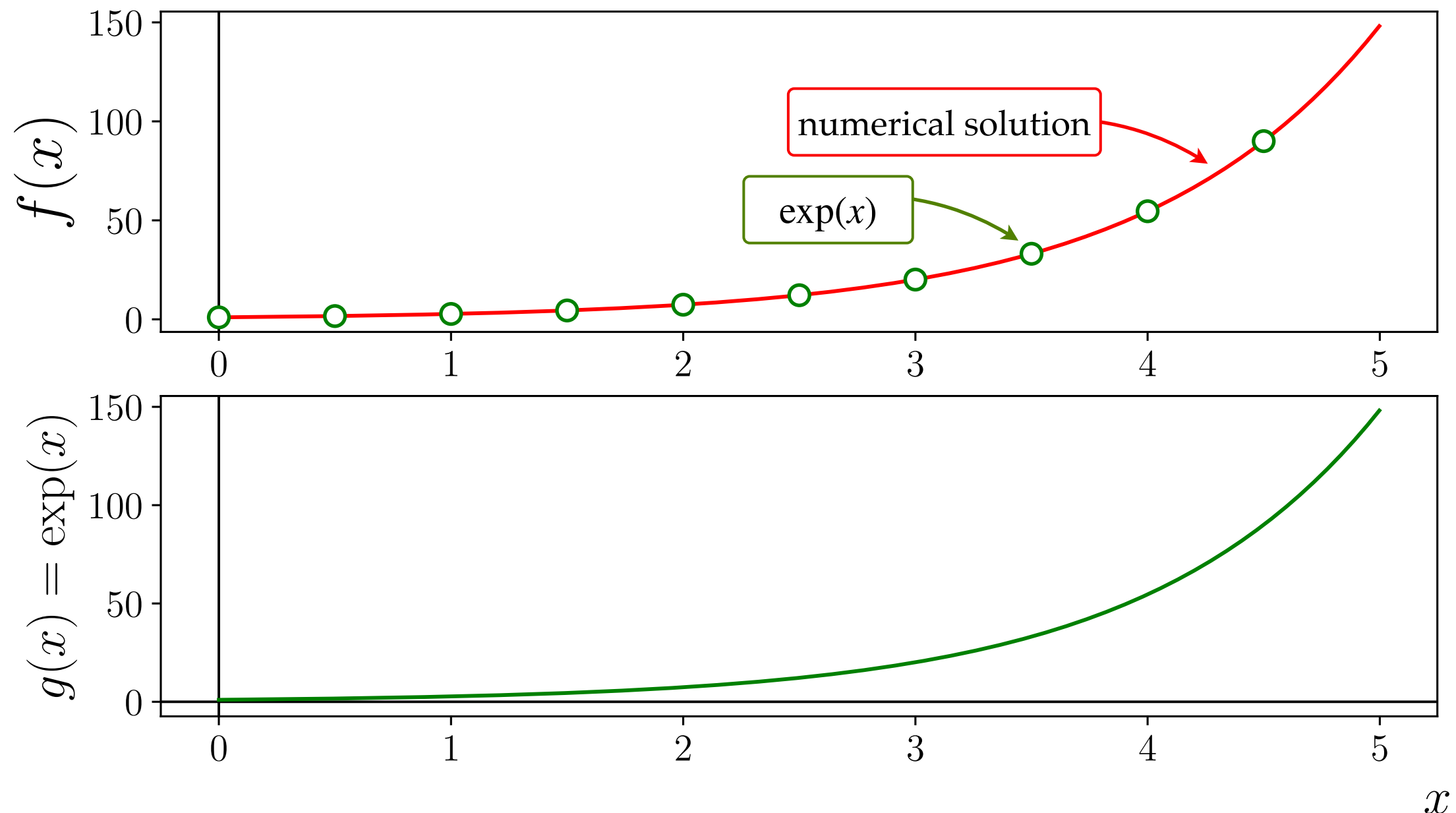
look inside my code 😎

Don't be scared! The code is very short, I just wrote lots of comment to explain its logic 😊

```
def diff_eq(g, f0, xs, a):  
    """  
    this solves the diff. eq. df/dx = g(x) by discretizing it  
    (f(x+a) - f(x-a))/2a = g(x)  
    f(x+a) = f(x-a) + 2*a*g(x)  
  
    or equivalently  
  
    f(x+2 a) = f(x) + 2*a*g(x+a)  
  
    we put the list of points into fs  
  
    we initiate this with  
    the initial value of f, which is f0 = f(xs[0])  
    """  
    fs=[f0]  
  
    "note in this loop, we skip over the first element"  
    for i0 in range(1,len(xs)):  
        "we grab the previous term in the list"  
        f0 = fs[-1]  
  
        """  
        we need the derivative at x+a,  
        since xs[i0] = x+2*a  
        we can use the fact that x+a = xs[i0]-a  
        """  
        df = g(xs[i0]-a)*2*a  
  
        "we add these together"  
        fs.append(f0+ df)  
  
    return fs
```

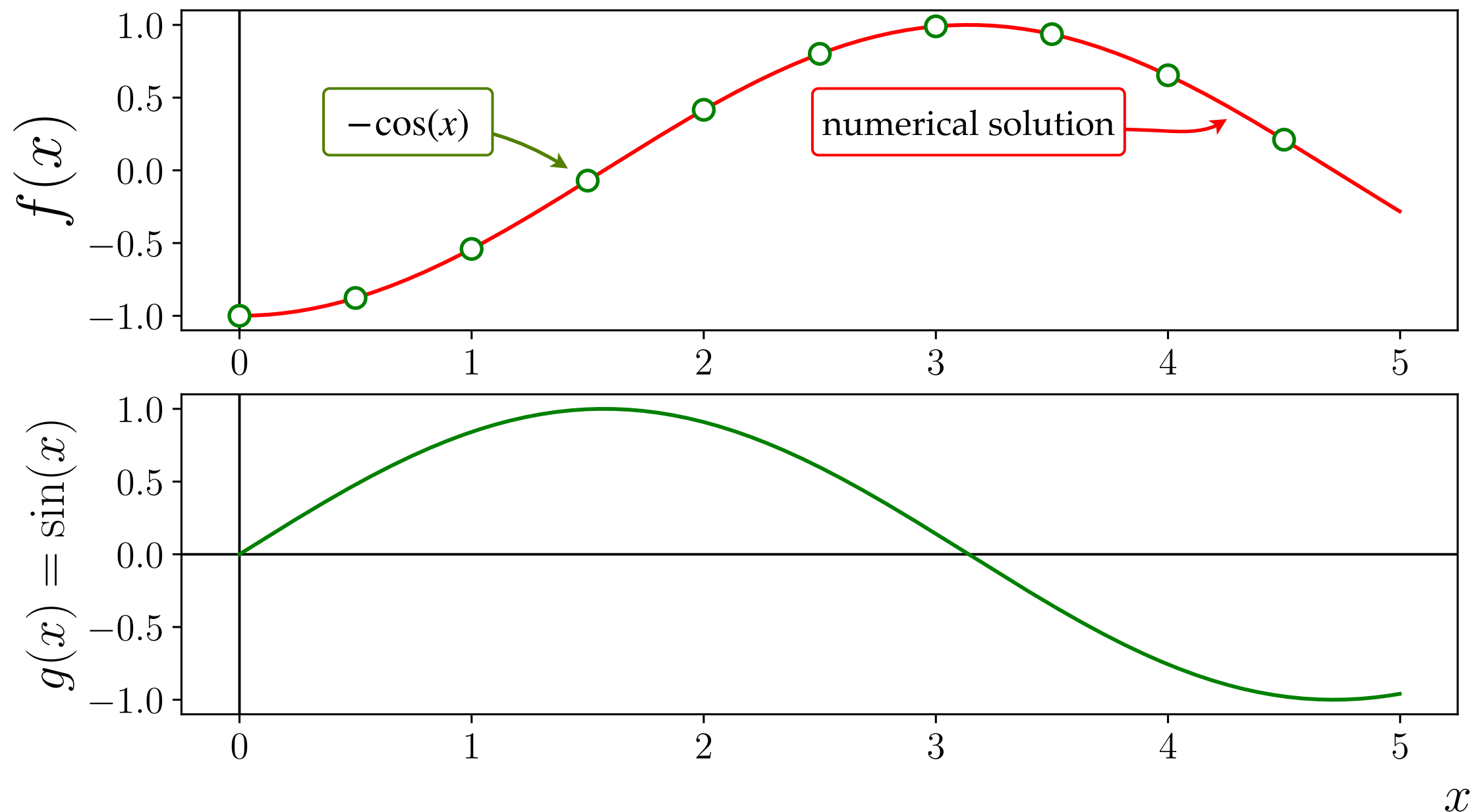
EXERCISE #5 - PUT YOUR CODE TO A TEST BY SOLVING FOR $f(x)$

Solve for $f(x)$ in this differential equation $\frac{df(x)}{dx} = g(x)$, by fixing $a = 10^{-3}$ and $x = [0,5)$, when $f(0) = 1$ and $g(x) = \exp(x)$.



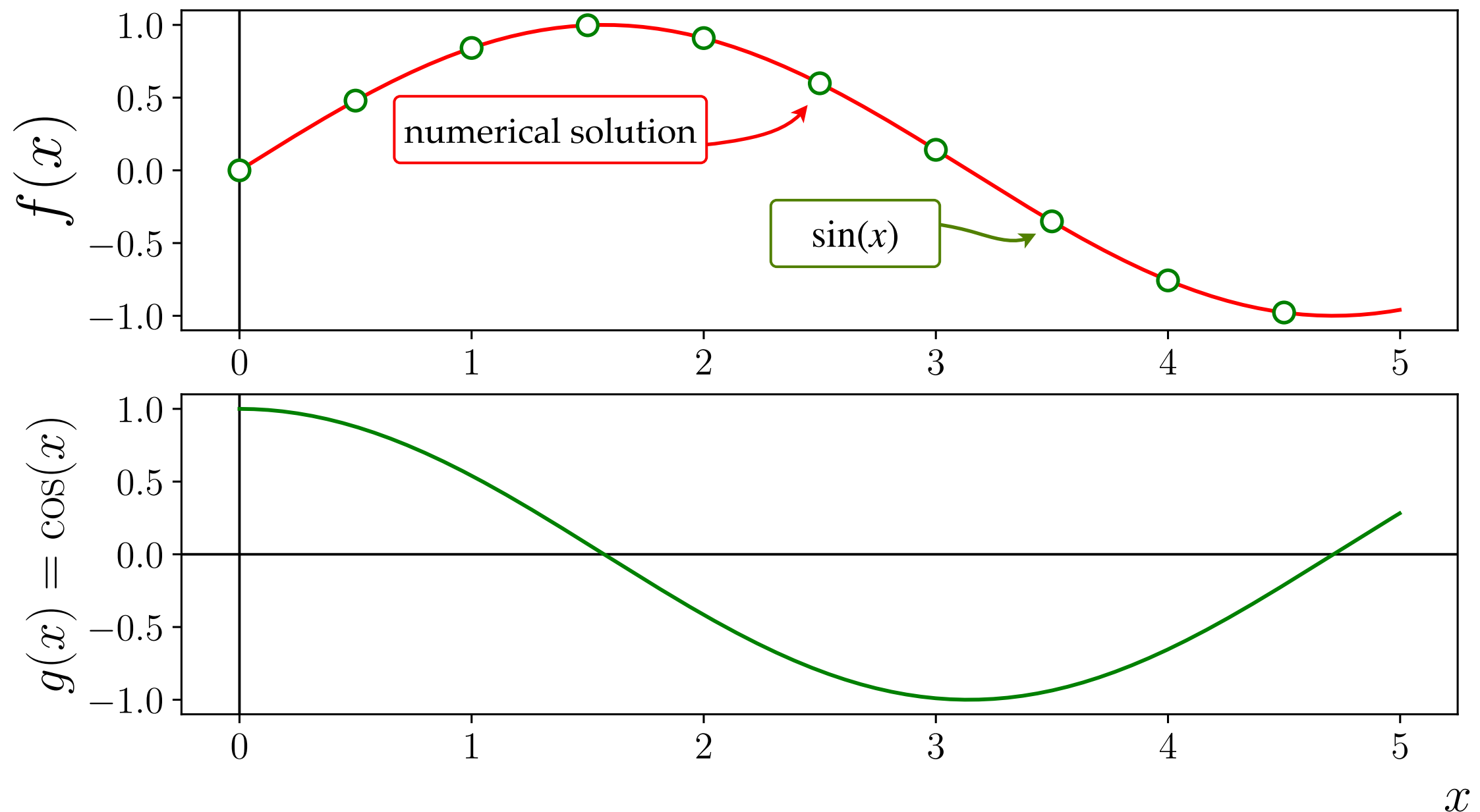
EXERCISE #5 - PUT YOUR CODE TO A TEST BY SOLVING FOR $f(x)$

Solve for $f(x)$ in this differential equation $\frac{df(x)}{dx} = g(x)$, by fixing $a = 10^{-3}$ and $x = [0,5)$, when $f(0) = -1$ and $g(x) = \sin(x)$.



EXERCISE #5 - PUT YOUR CODE TO A TEST BY SOLVING FOR $f(x)$

Solve for $f(x)$ in this differential equation $\frac{df(x)}{dx} = g(x)$, by fixing $a = 10^{-3}$ and $x = [0,5)$, when $f(0) = 0$ and $g(x) = \cos(x)$.

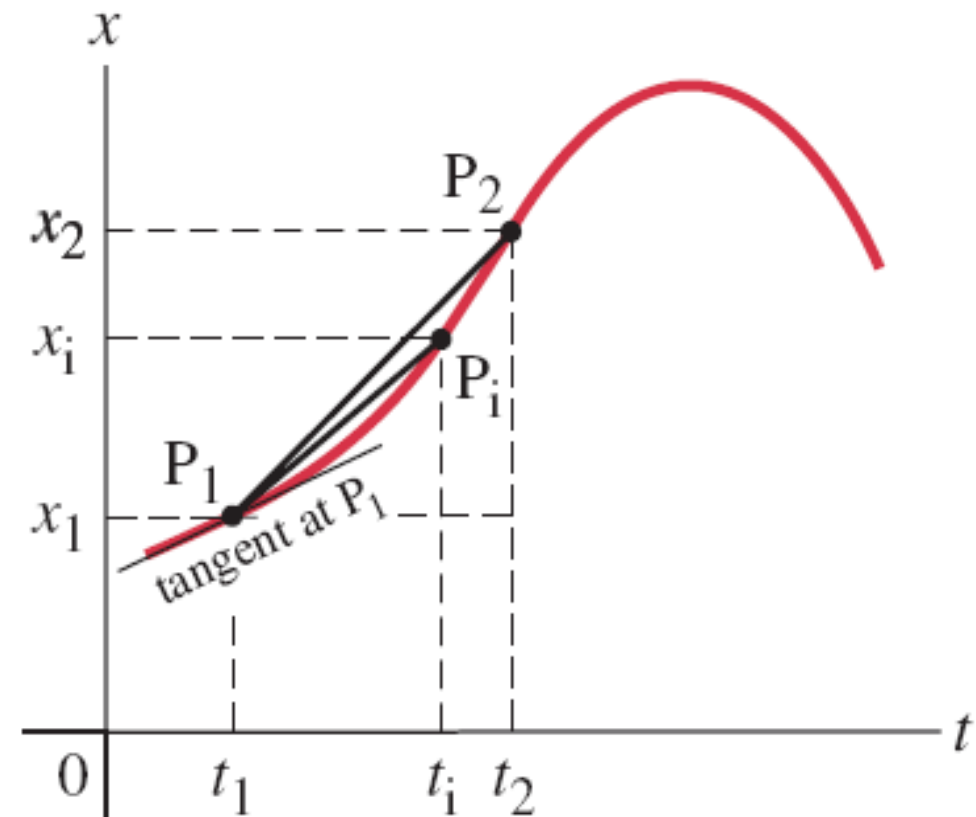
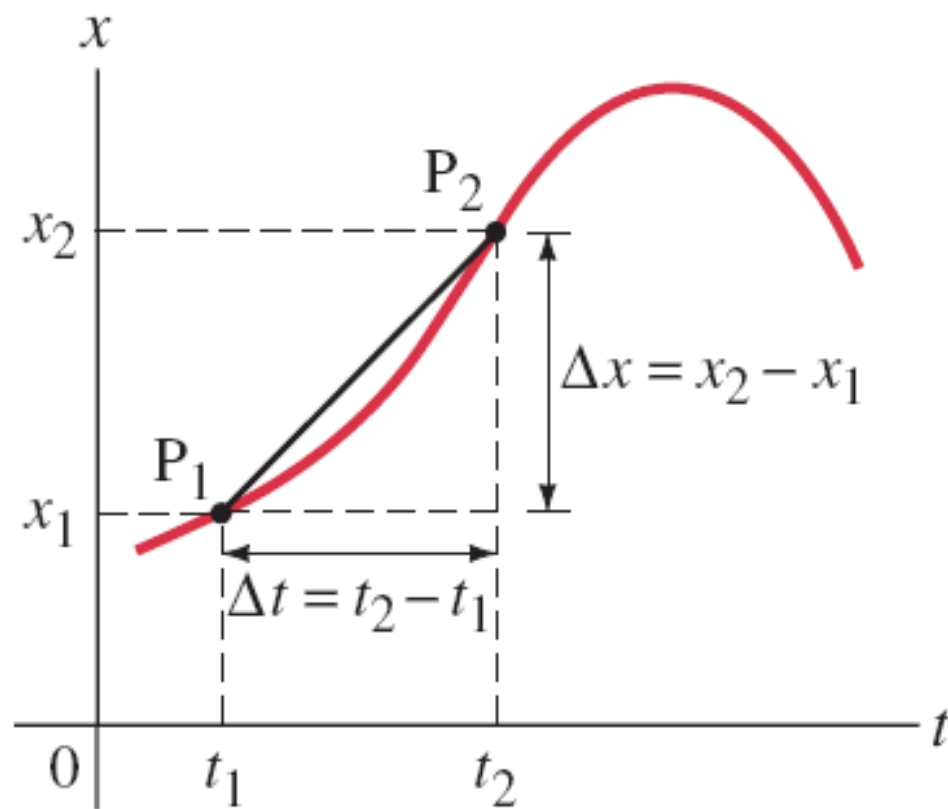


SECOND ORDER DIFFERENTIAL EQUATION



VELOCITY - DERIVATIVE OF POSITION

Instantaneous velocity is a vector defined as “how fast” a particle is moving at a given instant.



$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1}$$



$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

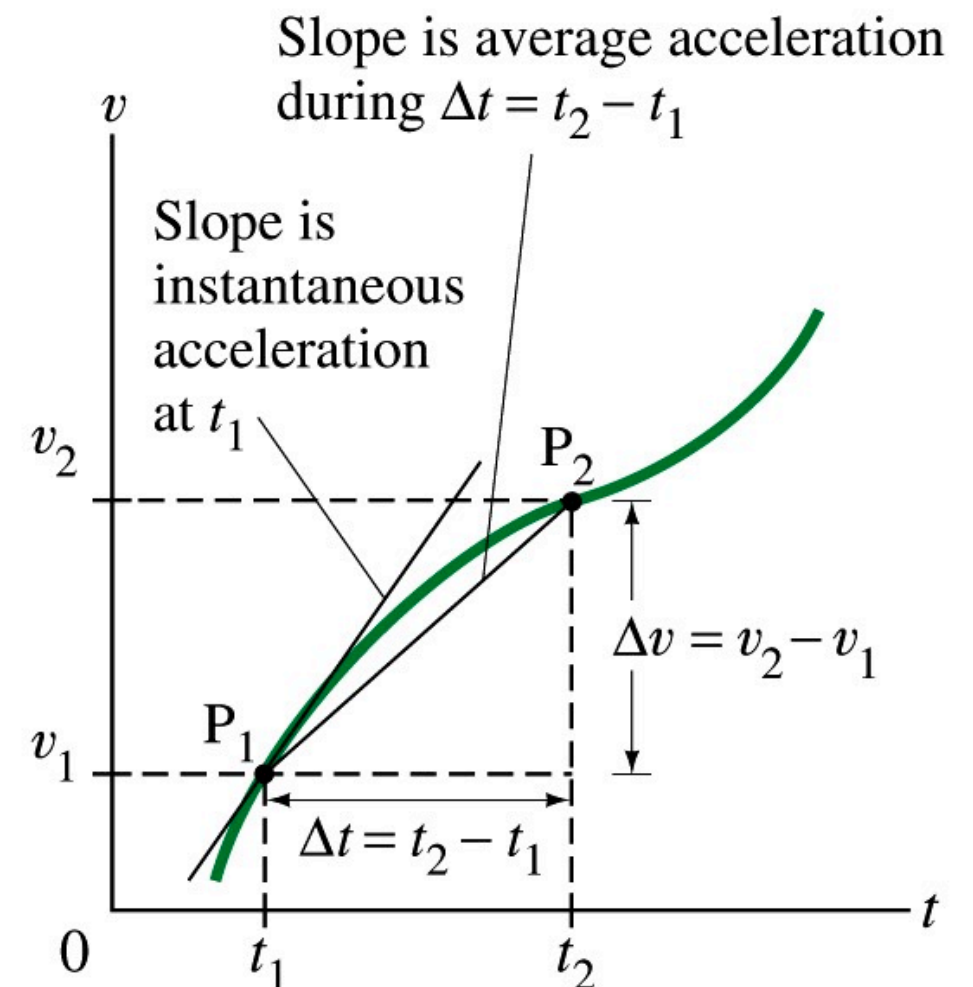
Instantaneous velocity is the tangent to the curve at any point.

ACCELERATION - DERIVATIVE OF VELOCITY

Instantaneous acceleration is the average acceleration in the limit as the time interval becomes infinitesimally short.

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Instantaneous acceleration is the slope of the tangent to the v vs. t curve at that time.



PROJECTILE MOTION

Remember the projectile motion problem we discussed on the first week?

[Where did these equations come from?]

Position as a function of time

$$x(t) = x_0 + v_0^x t + \frac{a_x}{2} t^2$$
$$y(t) = y_0 + v_0^y t + \frac{a_y}{2} t^2$$

Video Stream Archives



REYES: Remote Experience for Young Engineers and Scientists

Session Title: Python4Physics

Video Archives

Please click on the thumbnail to play

Via CiscoWebex

2D Kinematics

Free-fall parabolic motion:

$$y(t) = y_0 + v_0^y t - \frac{g}{2} t^2$$

$$x(t) = x_0 + v_0^x t$$

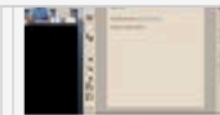
$$g = 9.8 \text{ m/s}^2 = 32.0 \text{ ft/s}^2$$

- 2D kinematics with constant acceleration: Parabolic

TR

Ted Rogers

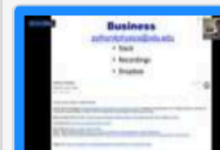
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Duration: 02:01:25
Date: 07-07-2020
Caption:



Python 4 Physics Course
Duration: 02:01:17
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Python 4 Physics Course
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Python 4 Physics Course
Duration: 01:57:38
Date: 06-25-2020
Caption:



Python 4 Physics Course
Duration: 01:58:31
Date: 06-23-2020
Caption:

44:39 / 2:00:10



1x



Event Information

Need Help?

Your Feedback?

PROJECTILE MOTION

Remember the projectile motion problem we discussed in the first week?

[Where did these equations come from?]

Position as a function of time

$$\begin{aligned}x(t) &= x_0 + v_0^x t + \frac{a_x}{2} t^2 \\ y(t) &= y_0 + v_0^y t + \frac{a_y}{2} t^2\end{aligned}$$

They are solutions to the second order differential equation for the each coordinate:

$$\begin{aligned}a_x(t) &= \frac{d^2}{dt^2} x(t), \\ a_y(t) &= \frac{d^2}{dt^2} y(t).\end{aligned}$$

You get the solutions above, if both $a_x(t)$ and $a_y(t)$ are both time independent. In which case the equations simplify

$$\begin{aligned}a_x &= \frac{d^2}{dt^2} x(t), \\ a_y &= \frac{d^2}{dt^2} y(t).\end{aligned}$$

SOLVING NUMERICALLY SECOND ORDER DIFFERENTIAL EQUATIONS - PART1

Having solved the simplest class of differential equations, let us consider the second easiest. For convenience we will change the independent variable to t , but we can still use the previous finding by replacing t with x .

We consider the second derivative of the function $f(t)$, and we set it equal to another function $h(t)$.

$$\frac{d^2}{dt^2}f(t) = h(t).$$

This might seem challenging, but it is because relatively easy if we define first the single derivative of $f(t)$ to be equal to $g(t)$

$$\frac{d}{dt}f(t) = g(t).$$

Then the first equation can be rewritten as

$$\frac{d}{dt} \left(\frac{d}{dt}f(t) \right) = \frac{d}{dt}g(t) = h(t).$$

Then we see that we have replaced the second order differential equation, with two single order differential equations

$$\frac{d}{dt}f(t) = g(t), \quad \text{and} \quad \frac{d}{dt}g(t) = h(t).$$

SOLVING NUMERICALLY SECOND ORDER DIFFERENTIAL EQUATIONS - PART2

We have simplified the problem, at the cost of having two unknown functions: $f(t)$ and $g(t)$. Fortunately, we have two equations and two unknowns, so we can indeed solve for the unknowns!

We can solve these two differential equations numerically by introducing a finite values of a as before. We get two coupled equations:

$$\begin{aligned}\frac{g(t+a) - g(t-a)}{2a} &= h(t), \\ \frac{f(t+a) - f(t-a)}{2a} &= g(t).\end{aligned}$$

Solving for $g(t+a)$ and $f(t+a)$

$$\begin{aligned}g(t+a) &= g(t-a) + h(t)2a, \\ f(t+a) &= f(t-a) + g(t)2a.\end{aligned}$$

As before, we can shift the argument by a to rewrite these solutions as

$$\begin{aligned}g(t+2a) &= g(t) + h(t+a)2a, \\ f(t+2a) &= f(t) + g(t+a)2a.\end{aligned}$$

SOLVING NUMERICALLY SECOND ORDER DIFFERENTIAL EQUATIONS - PART3

Note, in the equation we need to calculate $g(t + a)$, but we never quite calculate this. Instead, we calculate $g(t + 2a)$ and $g(t)$. For sufficiently small values of a , you can approximate this by $g(t + 2a)$, and the error you would introduce would be linear in a . This means that the error would vanish as you take the $a \rightarrow 0$ limit. Alternatively you can use the following average, and the error would be quadratic in a :

$$g(t + a) = \frac{g(t + 2a) + g(t)}{2} + \mathcal{O}(a^2),$$

Big O notation: it tells you how the error behaves in the asymptotic limit, which in this case refers to the $a \rightarrow 0$ limit.

🧐 Proving that the approximation scales like $\mathcal{O}(a^2)$ can be done by Taylor expanding both sides of the equality.

Now you can write similar code to before to loop over all values of both $f(t)$ and $g(t)$. Note, for this we need two initial values at $x = 0$: $f(0)$ and $g(0)$.

EXERCISE #6 - WRITE CODE FOR $x(t)$

First, write code for the expected behavior of $x(t)$ and $v(t)$ as a function of time.

Position as a function of time	$x(t) = x_0 + v_0^x t + \frac{a_x}{2} t^2$ $y(t) = y_0 + v_0^y t + \frac{a_y}{2} t^2$
Velocity as a function of time	$v_x(t) = x'(t) = v_0^x + a_x t$ $v_y(t) = y'(t) = v_0^y + a_y t$

```
def x_func(t, x0, v0, a0):  
    x1 = v0 * t  
    x2 = a0 * pow(t,2)/2.0  
    return x0 + x1 + x2  
  
def v_func(t, v0, a0):  
    return v0 + a0 * t
```

look inside my code 😎

EXERCISE #7 - WRITE CODE FOR $f(t)$, $g(t)$ GIVEN $h(t)$

Use these iterative equations

$$g(t + 2a) = g(t) + h(t + a)2a,$$

$$f(t + 2a) = f(t) + g(t + a)2a.$$

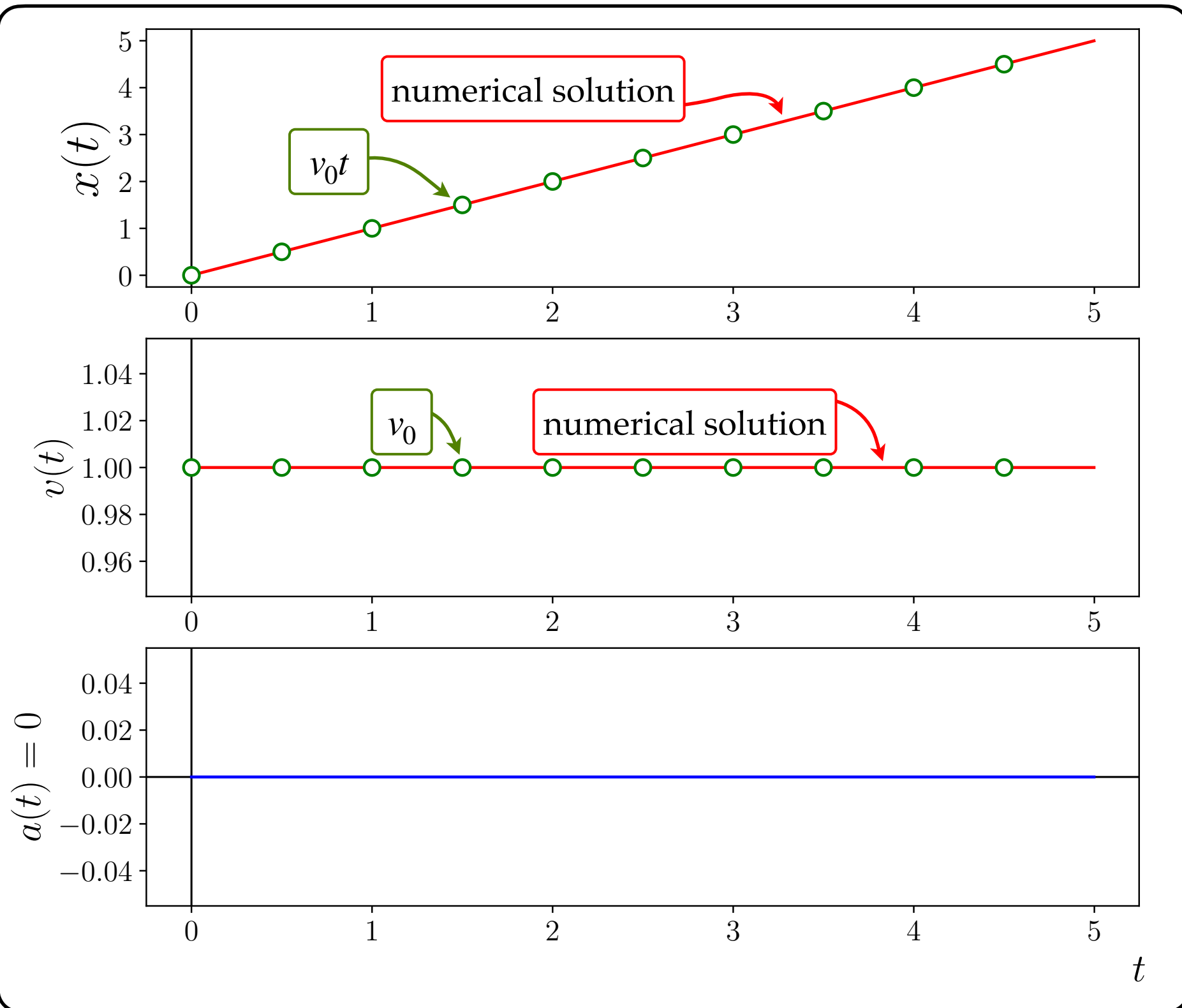
look inside my code 😎

Don't be scared! The code is very short, I just wrote lots of comment to explain its logic 🧐

```
def second_diff_eq(h, f0, g0, ts, a):  
    """  
    this solves the diff. eq.  $d^2f/dx^2 = h(x)$  by discretizing  
    re-writing it as a coupled linear equations  
  
     $dg/dt = h(t)$   
     $df/dt = g(t)$   
  
     $g(t + 2*a) = g(t) + 2 * a * h(t + a)$   
     $f(t + 2*a) = f(t) + 2 * a * g(t + a)$   
  
    we put the list of points into fs, gs  
  
    we initiate this with  
    the initial value of f, which is  $f0 = f(ts[0])$   
    the initial value of g, which is  $g0 = g(ts[0])$   
  
    note, we need to calculate  $g(t + a)$ , but we have  $g(t + 2*a)$  and  $g(t)$   
    we will estimate this with the average of these two, which we will call  
  
     $\underline{gbar} = ( g(t + 2*a) + g(t) ) / 2$   
    """  
  
    fs=[f0]  
    gs=[g0]  
  
    "note in this loop, we skip over the first element"  
    for i0 in range(1,len(ts)):  
        "we start with  $g(t)$ : we grab the previous term in the list"  
        g0 = gs[-1]  
        "the shift in g"  
        dg = h(ts[i0] + a) * 2*a  
  
        gs.append( g0 + dg )  
  
        "next we calculate  $f(t)$ : we grab the previous term in the list"  
        f0 = fs[-1]  
        "we estimate  $g(t+a)$  with the average of the last two point in gs"  
        gbar = (gs[-1] + gs[-2]) / 2.0  
        df = gbar * 2 * a  
  
        "we add these together"  
        fs.append(f0 + df)  
  
    return fs, gs
```

EXERCISE #8 - REPRODUCE THESE PLOTS

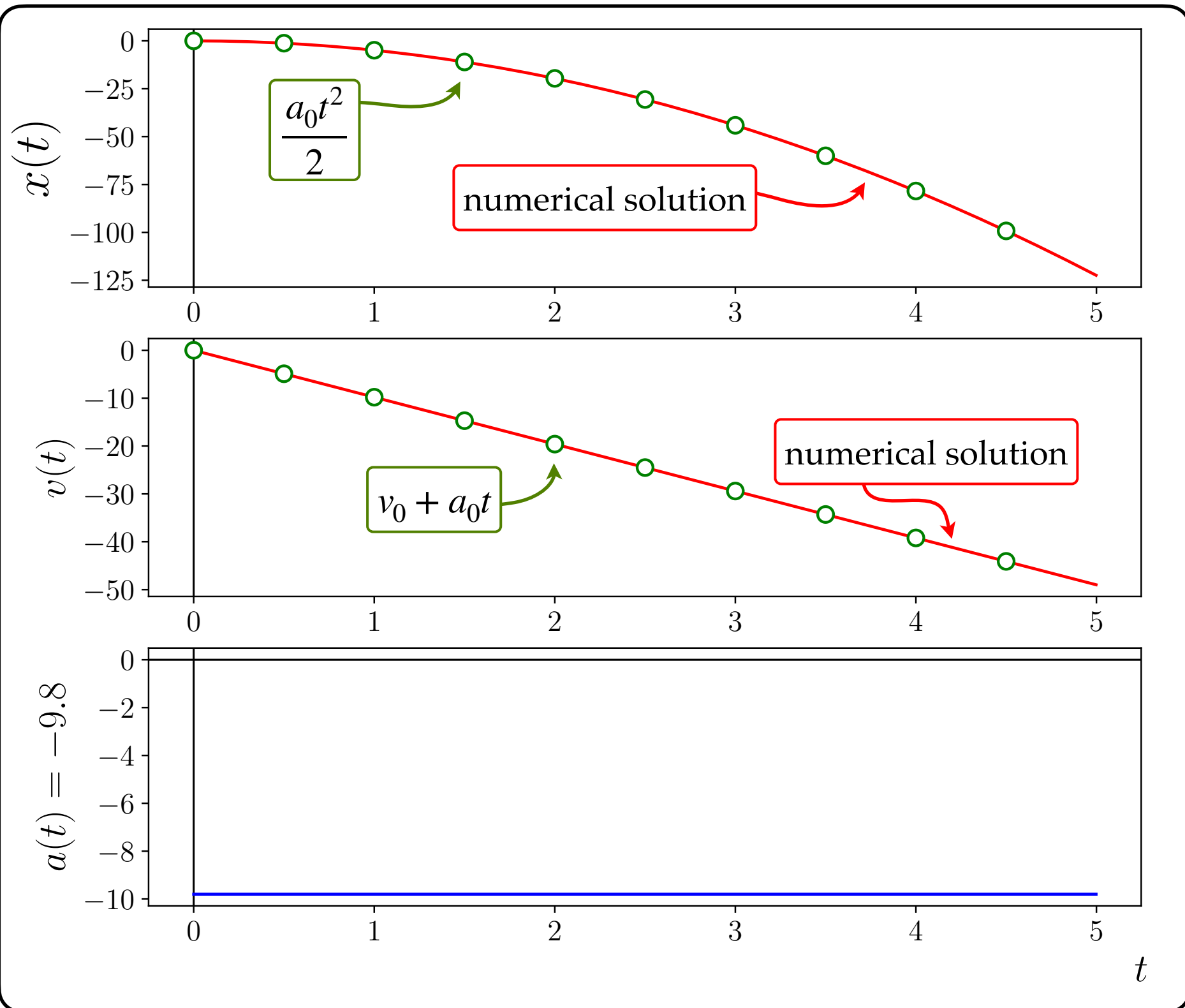
Test your code and test the kinematic equations obtained assuming a constant acceleration.
For $(x_0, v_0) = (0, 1)$ and $a(t) = 0$.



$$x(t) = x_0 + v_0^x t + \frac{a_x}{2} t^2$$

EXERCISE #8 - REPRODUCE THESE PLOTS

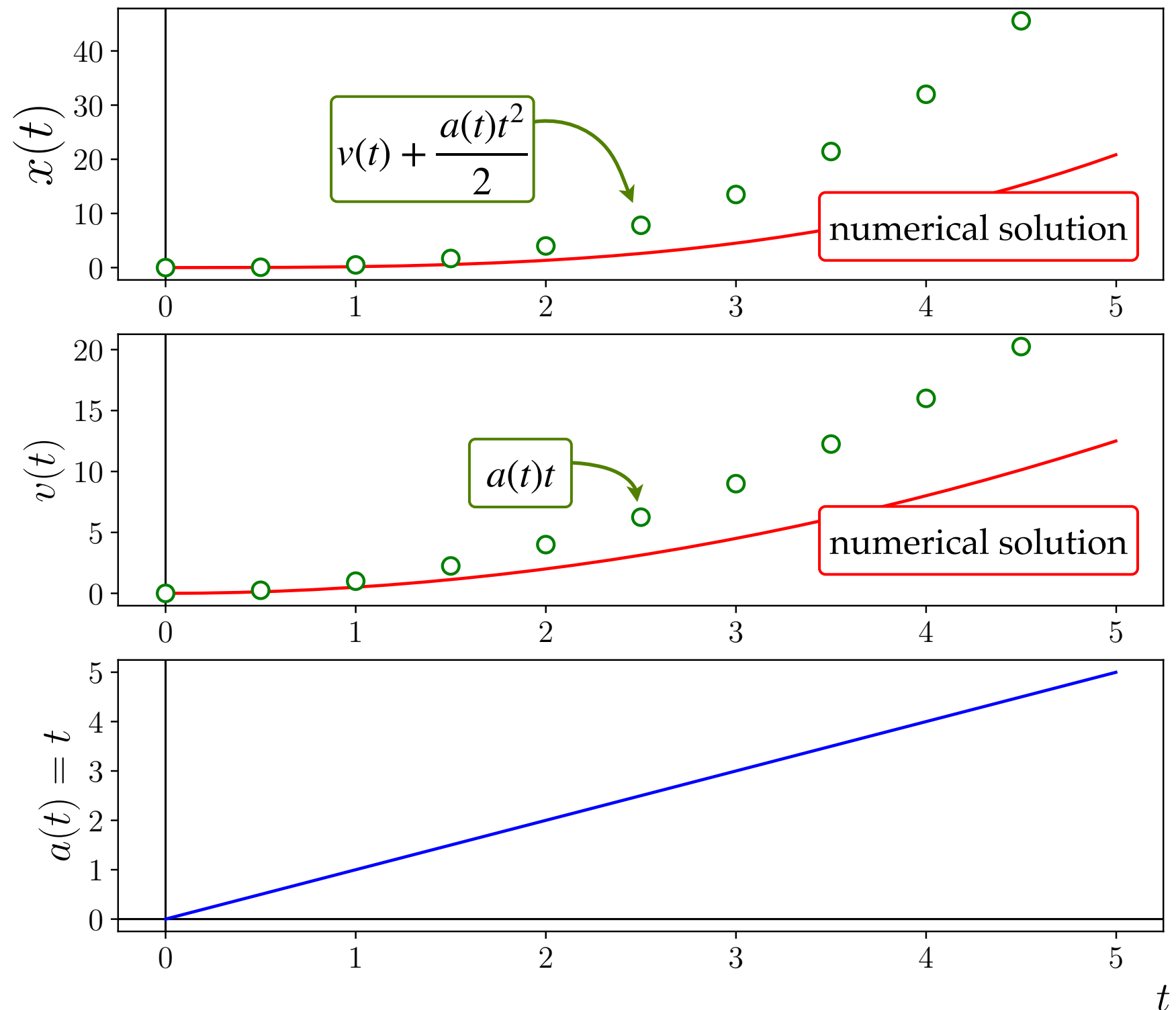
Test your code and test the kinematic equations obtained assuming a constant acceleration.
For $(x_0, v_0) = (0,0)$ and $a(t) = -9.8$.



$$x(t) = x_0 + v_0^x t + \frac{a_x}{2} t^2$$

EXERCISE #8 - REPRODUCE THESE PLOTS

Test your code and test the kinematic equations obtained assuming a constant acceleration.
For $(x_0, v_0) = (0,0)$ and $a(t) = t$.



$$x(t) = x_0 + v_0^x t + \frac{a_x}{2} t^2$$

SUMMARY - CONCEPTS COVERED

Physics concepts / equations

Position

$$\text{Velocity : } v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

$$\text{Acceleration : } a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

Calculus

$$\text{Derivatives: } \frac{d}{dx} f(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

Numerical solution to differential equations

$$\text{Equation: } \frac{d}{dx} f(x) = g(x) \quad \text{Solution: } f(x + 2a) = f(x) + g(x + a)2a$$

$$\text{Equation: } \frac{d^2}{dt^2} f(t) = h(t)$$

$$\text{Solution: } g(t + 2a) = g(t) + h(t + a)2a, \quad f(t + 2a) = f(t) + g(t + a)2a.$$

Python

Functions of function: `def f(g, x): return g(x)`